Lecture 10

Logistic Regression (1)

Introduction to Logistic Regression

EECS 189/289, Fall 2025 @ UC Berkeley
Joseph E. Gonzalez and Narges Norouzi

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Roadmap

- Classification Task
- Defining a New Model for Classification
- Linear Discriminant Functions
- Discriminative Probabilistic Models
- Sigmoid for Classification
- Logistic Regression Objective
- Regularization with Logistic Regression
- Logistic Regression Optimization
- Making Decisions

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Classification Task

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So Far ....

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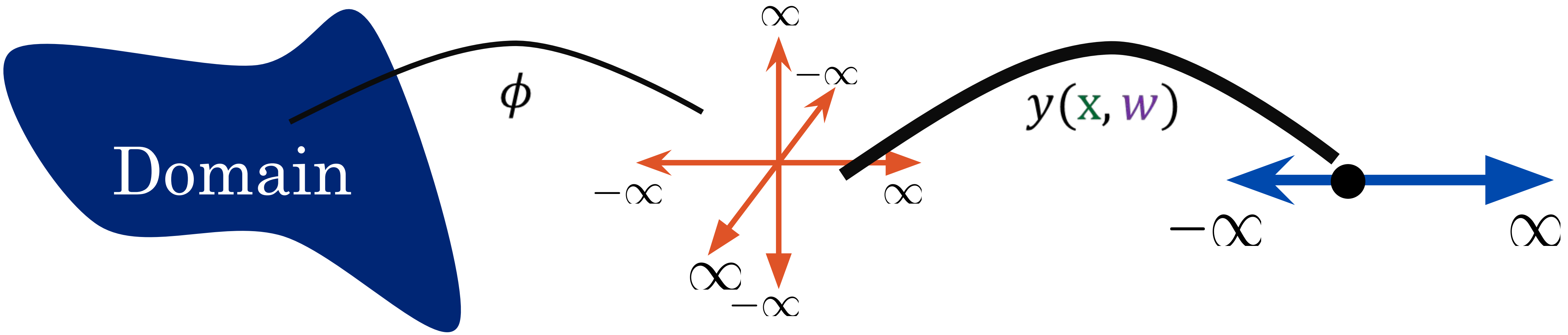

Squared Error

$$\underset{w}{\operatorname{argmin}} \frac{1}{2} \sum_{n=1}^{N} \left(t_n - w^T \phi(\mathrm{x}_n)\right)^2 + \frac{\lambda}{2} w^T w$$

Regularization

Classification Task Shift

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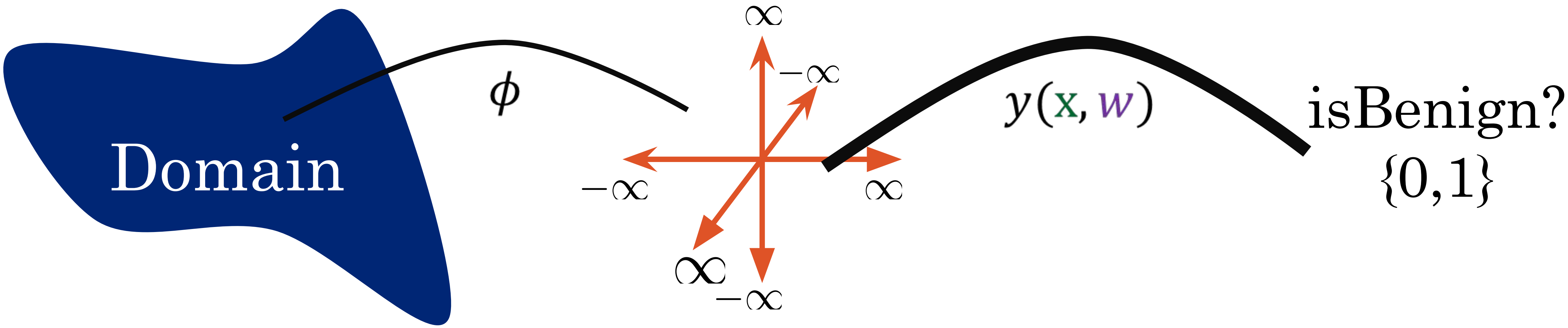

Kinds of Classification

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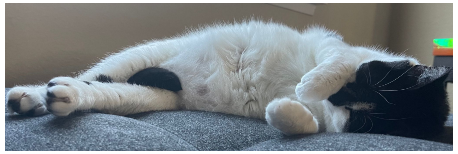
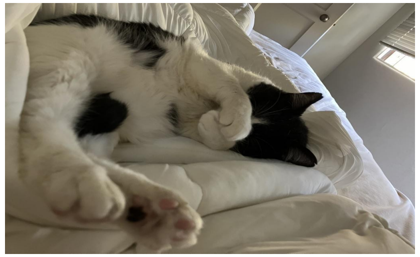
We want to predict some **categorical variable**, or **response**, y.

- **Binary** classification
  - Two classes
  - **Responses** (y) are either 0 or 1
- **Multiclass** classification
  - Many classes
  - Image labeling and sentiment analysis (positive, negative, or neutral).
- **Structured prediction** tasks
  - Predicting a structured object instead of a discrete class
  - Examples: Translation and ChatGPT.

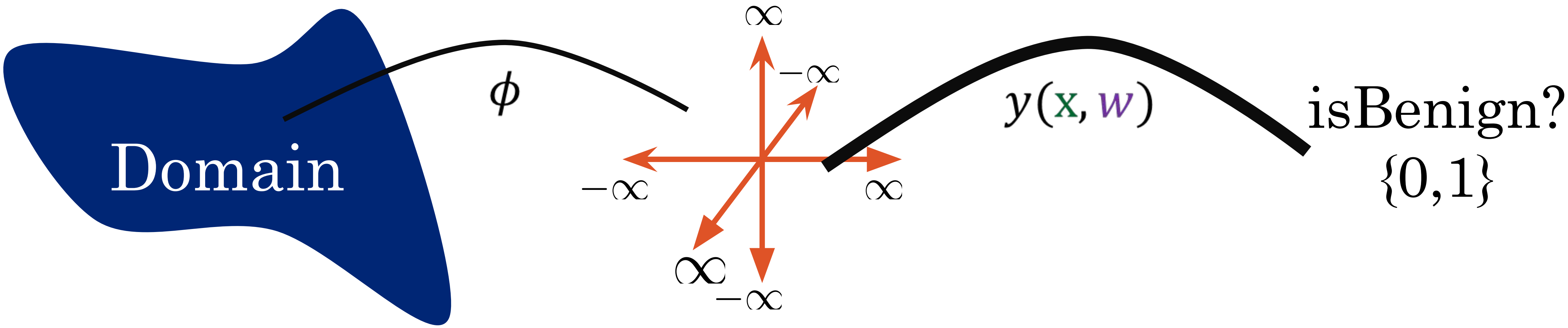
win or lose

disease or no disease





Classification



Can we just use least squares?

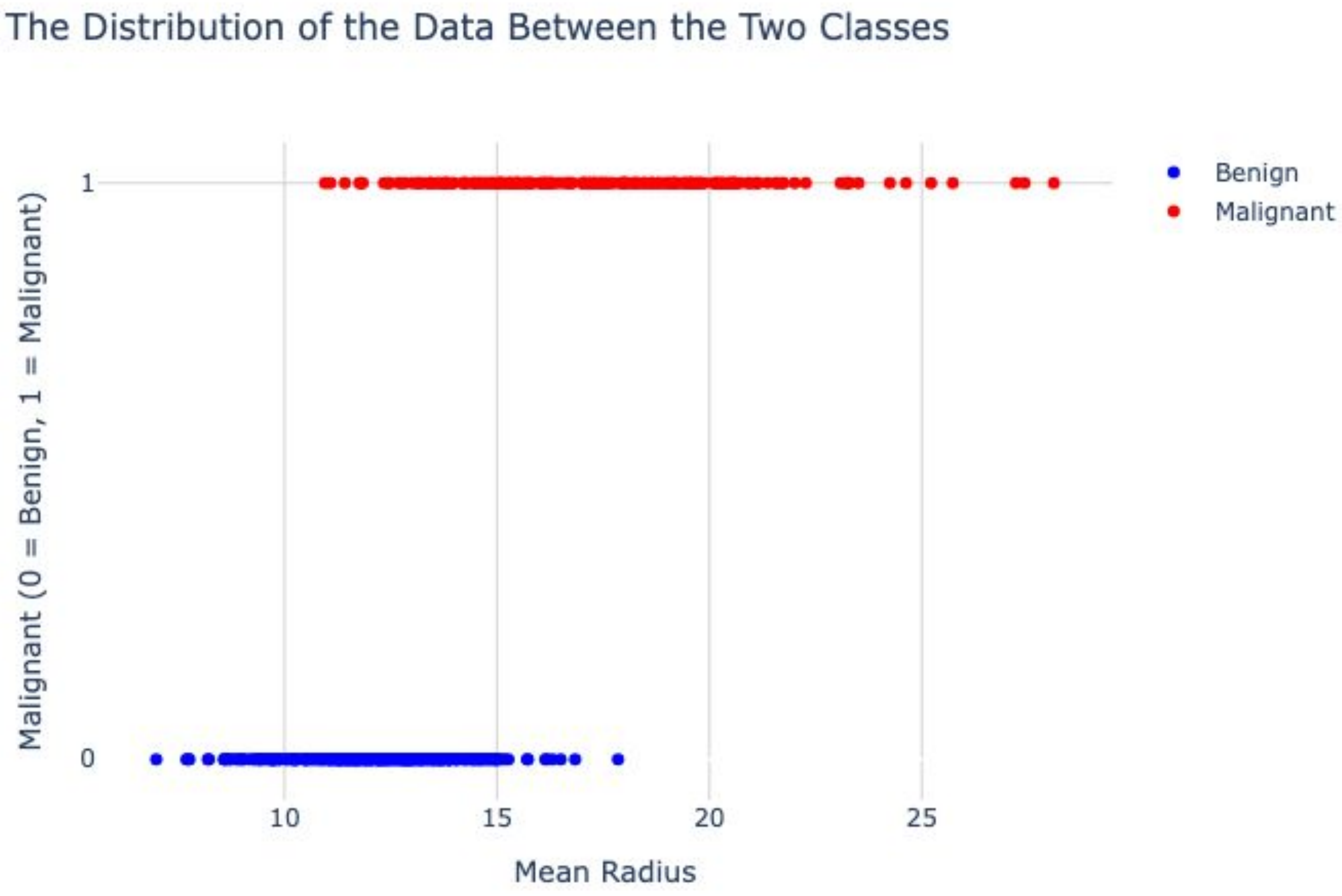
$$\underset{w}{\operatorname{argmin}} \frac{1}{2} \sum_{n=1}^{N} \left(t_n - w^T \phi(x_n)\right)^2 + \frac{\lambda}{2} w^T w$$

Squared Error

Regularization

Demo

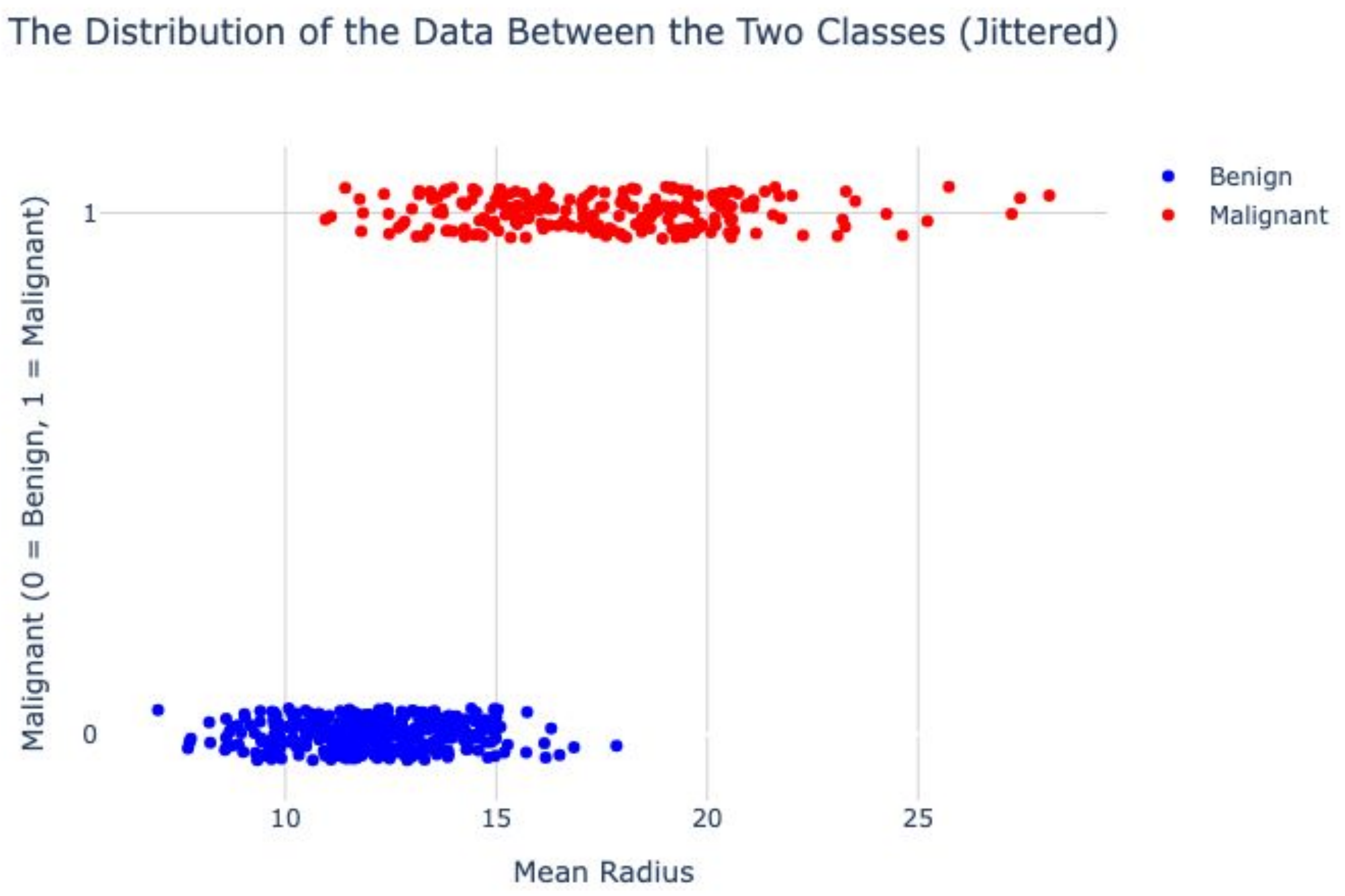
We see the distribution of the two classes in the figure. Y axis shows the class of the image (benign or malignant) and the X axis shows Mean Radius.



Demo

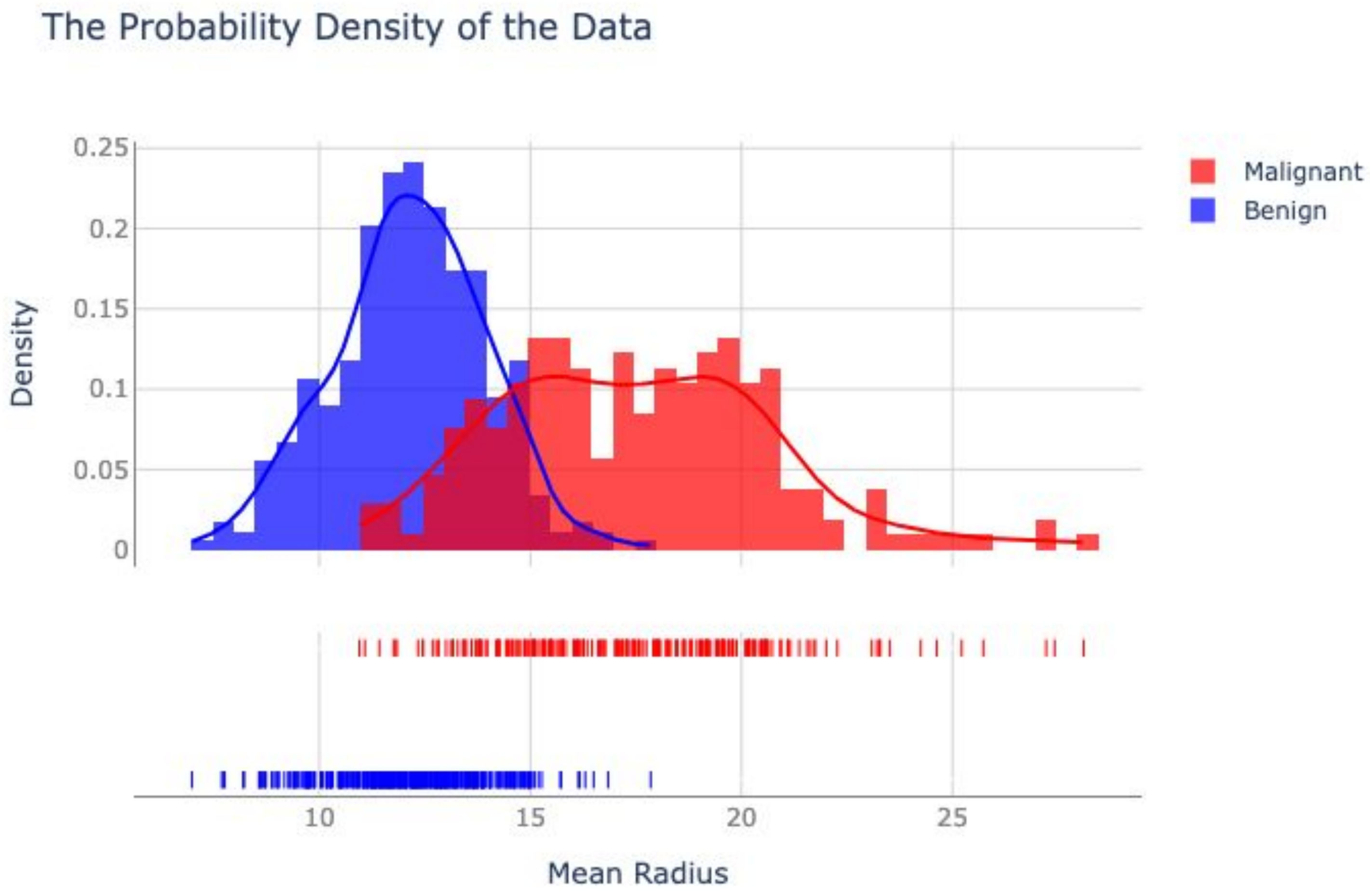
We usually add jittering to be able to see all data points clearly.

```
def jitter(data, amt=0.1):
return data +
    amt*np.random.rand(len(data)
    - amt/2.0
```



Demo

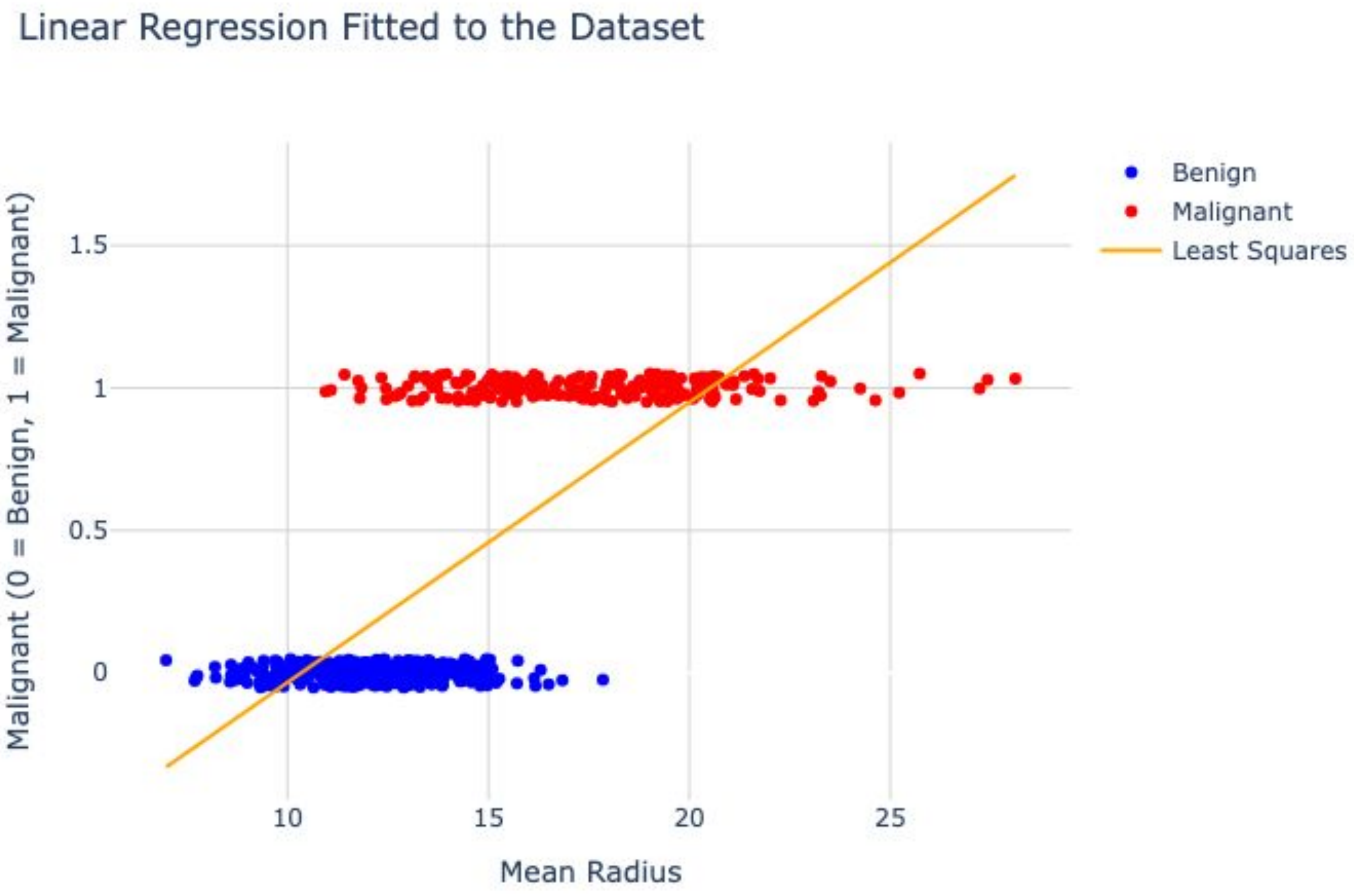
Here is the probability density of the data for both classes.



Demo

What if we test linear regression model for this task?

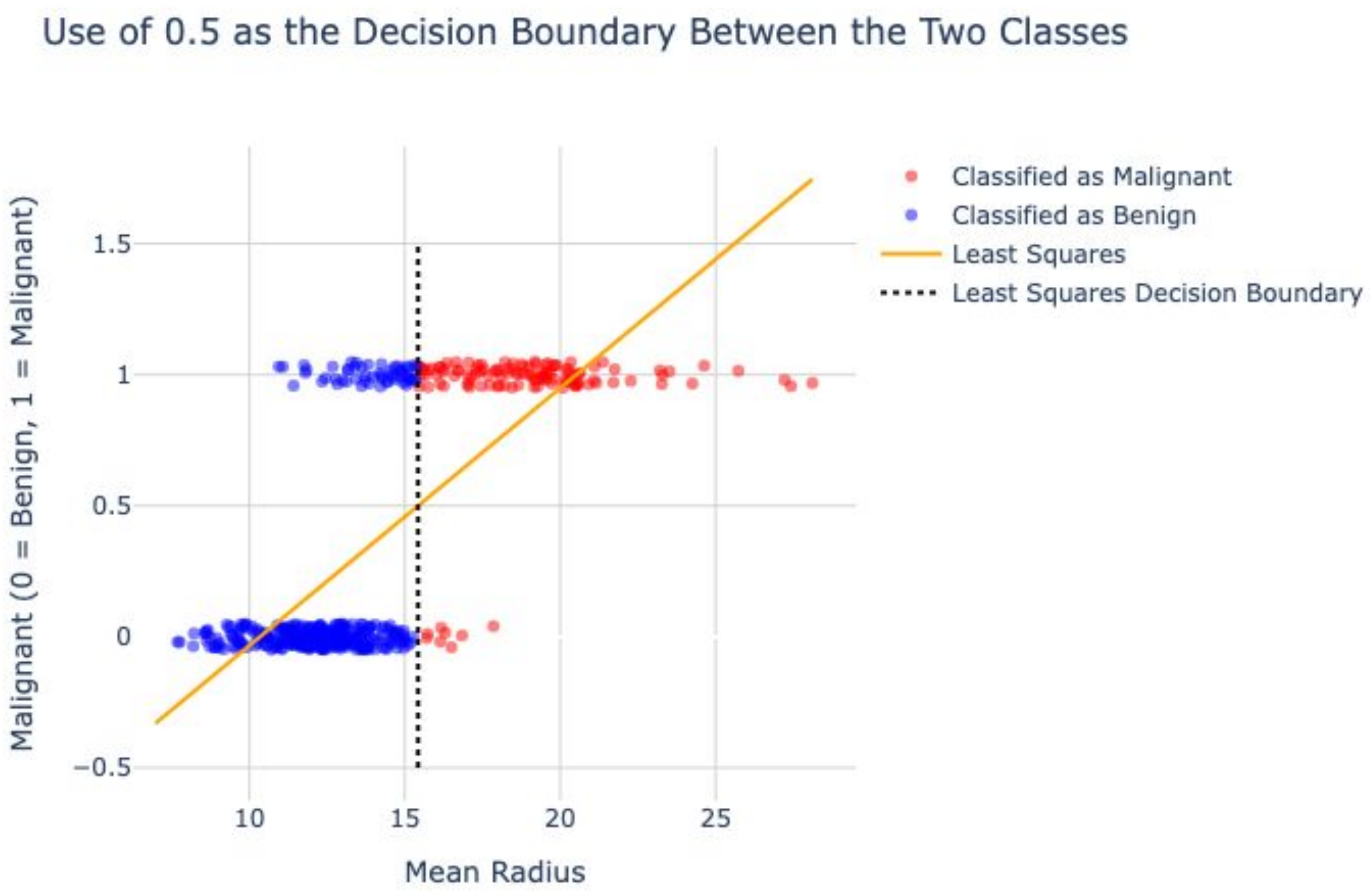
Here, the orange line is a regression line fit to the Mean Radius feature, mapping it to the scalar target values (0s and 1s).



Demo

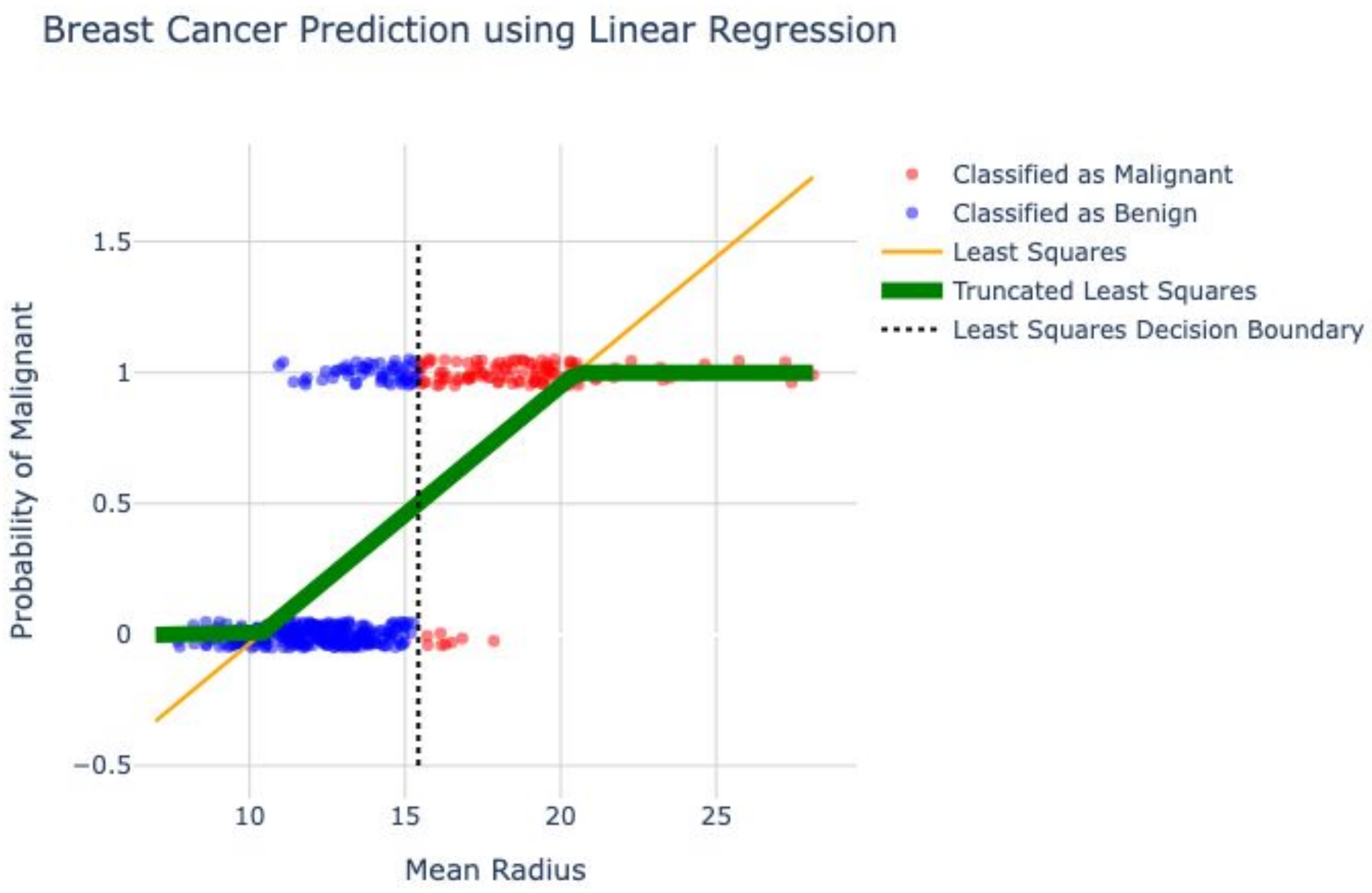
We needed a decision function (e.g. , $f(x) > 0.5$) ...

If we use the threshold of 0.5 between the two target values 0 and 1 and highlight all data points that are labeled as benign with blue and all malignant predictions with red, you will see that a decision boundary is being formed between the two classes with the black dashed line.



Demo

It is difficult to interpret a model that outputs values below 0 and above 1 so we capped the output between 0 and 1.



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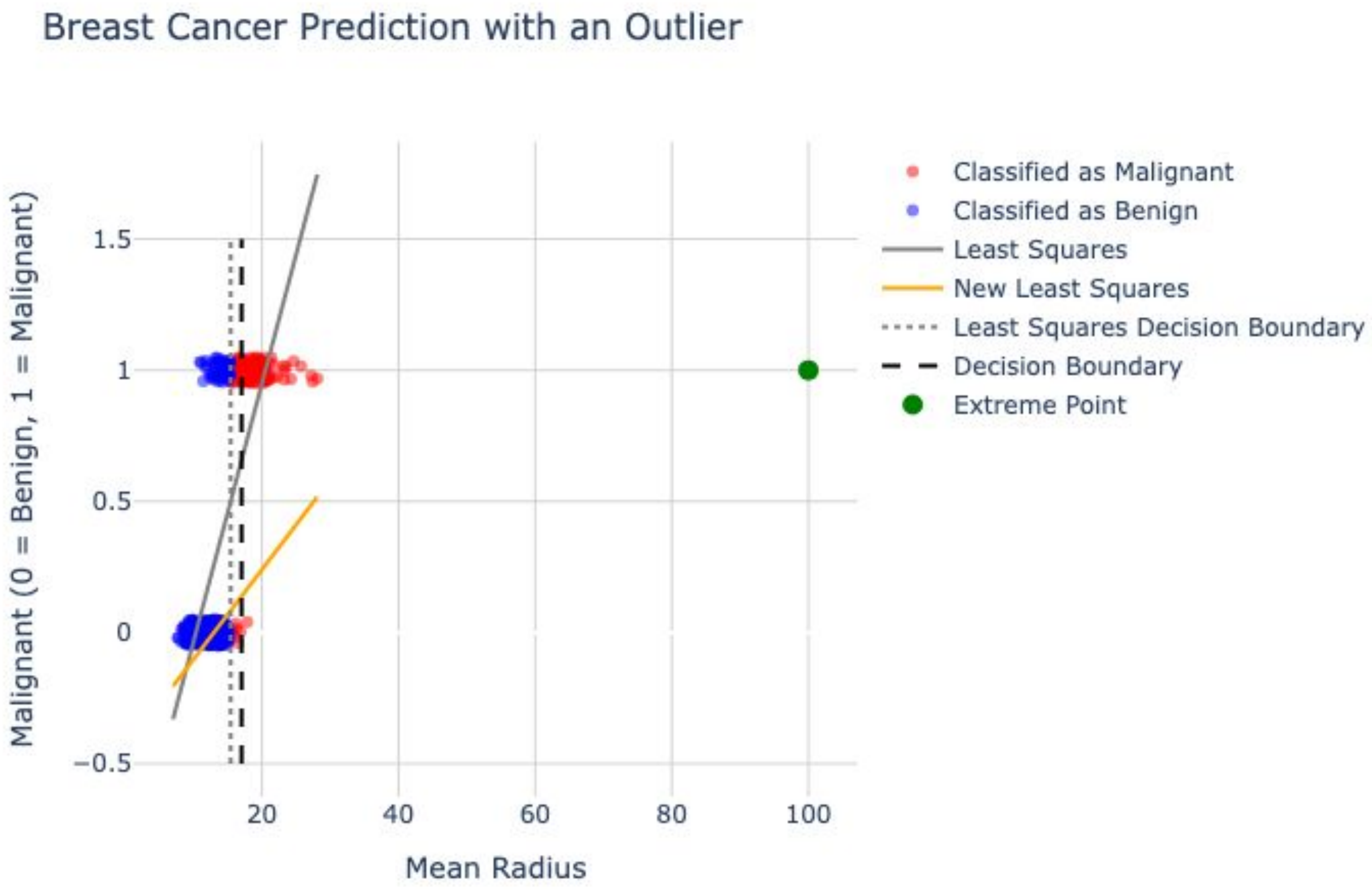
Demo

This model is also sensitive to outliers.



Classification

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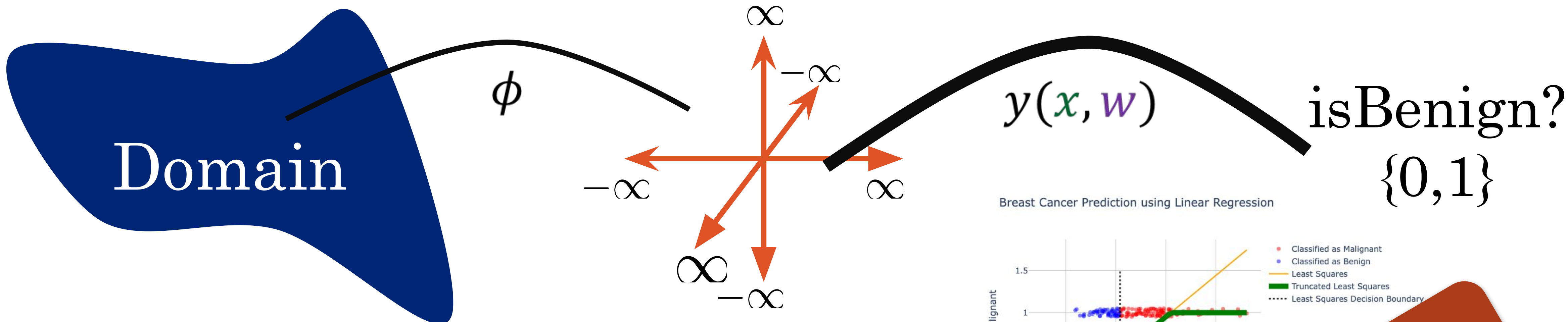

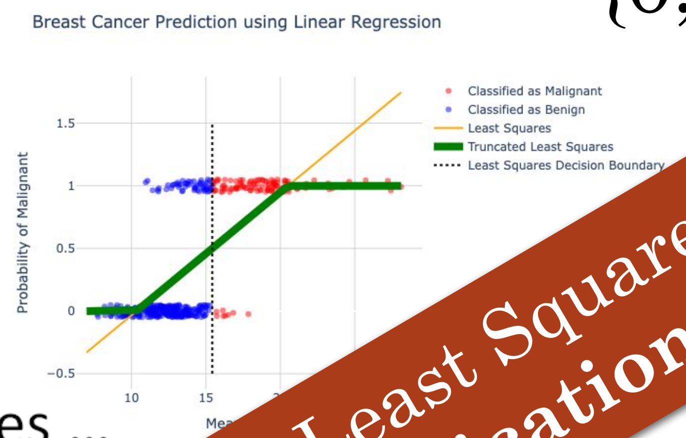

Can we just use least squares?

$$\underset{w}{\operatorname{argmin}} \frac{1}{2}\sum_{n=1}^{N}\left(t_n - w^T\phi(x_n)\right)^2 + \frac{\lambda}{2} w^T w$$

Squared Error

Regularization

- Yes ...
- Need ... tion (e.g.,
- ... interpret model ...
- Se... tive to outliers

Don't use Least Squares for Classification

Least squares assumes Gaussian prior, so it fails with binary data.

Takeaway!!!

Defining a New Model for Classification

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Classification Setup

- **Goal:** Map an input vector $\mathbf{x} \in \mathbb{R}^D$ to one of the K **discrete classes** C_k where $k = 1, 2, \ldots, K$.
  - $K = 2$: **Binary Classification**
  - $K > 2$: **Multi-class Classification**
- Input space is divided into decision regions. Boundaries are **decision boundaries** or **decision surfaces**.
  - $(D - 1)$-dimensional hyperplanes within the D-dimensional input space.

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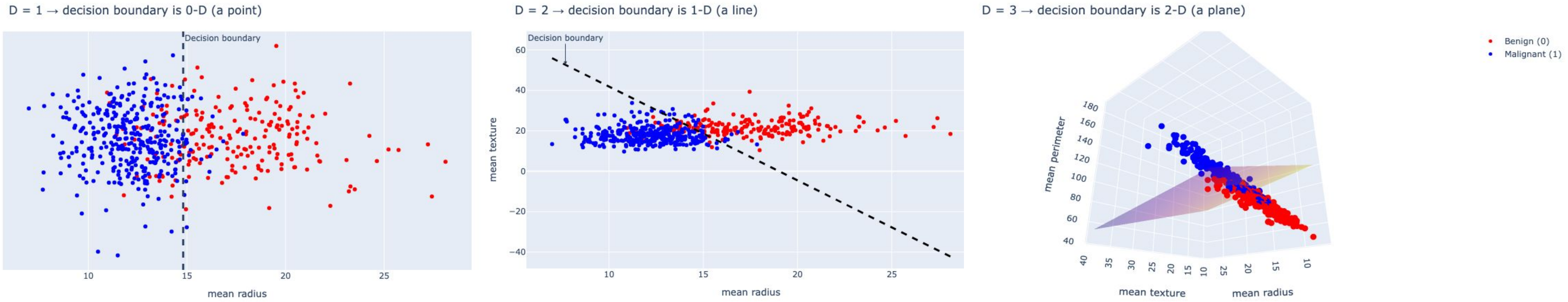
Classification Setup

- Input space is divided into decision regions. Boundaries are **decision boundaries** or **decision surfaces**.
  - $(D-1)$-dimensional hyperplanes within the D-dimensional input space.



- Dataset with classes that are separable by linear decision surfaces are said to be **linearly separable**.

3 Approaches in Solving Classification

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1. **Discriminant functions** (direct mapping): Direct assignment to a class.

Ex: SVM (margin-based view)

2. **Generative probabilistic models** (model conditional probability $p(\mathrm{x}|C_k)$ and prior $p(C_k)$ then apply Bayes).

Ex: Naïve Bayes, ChatGPT

$$p(C_k|\mathrm{x}) = \frac{p(\mathrm{x}|C_k)p(C_k)}{p(\mathrm{x})}$$

3. **Discriminative probabilistic models** (model conditional probability $p(C_k|\mathrm{x})$ directly).

Ex: Logistic Regression

Linear Discriminant Functions

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Discriminant Classification with Two Classes

- **Basic representation:**

  A linear function of the input vector so that $y(\mathrm{x}) = w^T\mathrm{x} + w_0$.

- **Decision rule:**

$$\text{Class Assignment} = \begin{cases} C_1 & \text{if } y(\mathrm{x}) \geq 0.5 \\ C_2 & \text{otherwise} \end{cases}$$

- **Decision boundary:**

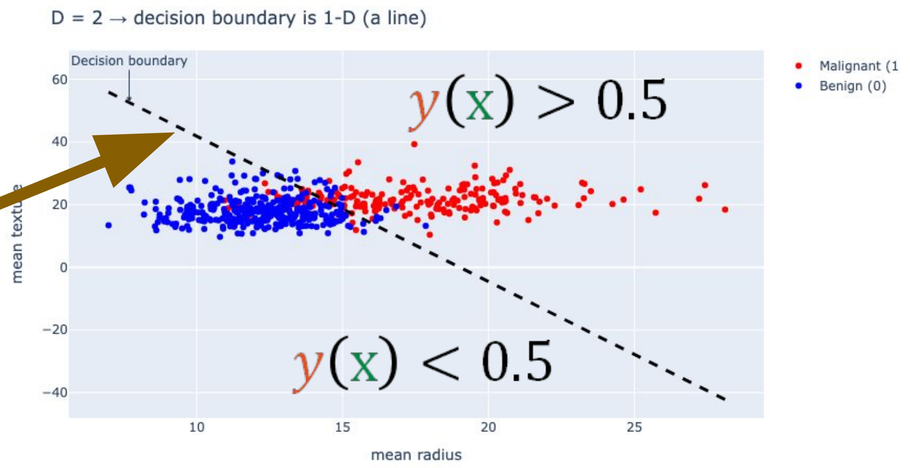
$$y(\mathrm{x}) = 0.5$$

Which is a $(D-1)$-dimensional surface.

$$y(\mathrm{x}) = 0.5$$

$$w^T\mathrm{x} + w_0 = 0.5$$

$$\mathrm{x}_{Mean\_radius} \times w_1 + \mathrm{x}_{Mean\_texture} \times w_2 + w_0 = 0.5$$



Geometry of Linear Discriminant Functions

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- For any two points on the line $\mathrm{x_A}$ and $\mathrm{x_B}$:

$$y(\mathrm{x}_A) = y(\mathrm{x}_B) = 0$$

$$w^T(\mathrm{x}_A - \mathrm{x}_B) = 0$$

- w is orthogonal to every vector lying within the decision surface/boundary.

w gives the orientation of the surface

Geometry of Linear Discriminant Functions

- For every point on the decision surface, $y(\mathrm{x}) = w^T\mathrm{x} + w_0 = 0$.
- The normal distance of the surface to the origin is:

$$\frac{w^T\mathrm{x}}{||w||} = -\frac{w_0}{||w||}$$

w_0 gives the location of the surface

If our model does not have an intercept term, the decision boundary must pass through the origin.

Discriminative Probabilistic Models

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Discriminative Probabilistic Models

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- Logistic regression is the most widely used **discriminative probabilistic model** for **classification.**

Logistic regression models the **log-odds** of class C_1 versus C_2 **as a linear model.**

$$\textbf{Odds Ratio} = \frac{p(C_1|\mathrm{x})}{p(C_2|\mathrm{x})} = \frac{p(C_1|\mathrm{x})}{1-p(C_1|\mathrm{x})}$$

$$\textbf{Log Odds} = \ln\left(\frac{p(C_1|\mathrm{x})}{p(C_2|\mathrm{x})}\right) = \ln\left(\frac{p(C_1|\mathrm{x})}{1-p(C_1|\mathrm{x})}\right) = \mathrm{w}^\top \mathrm{x}$$

We can solve for $p(C_1|\mathrm{x})$ to obtain the form of our model.

Deriving the Logistic Sigmoid Function

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We can solve for $p(C_1|\mathrm{x})$ to obtain the form of our model.

$$\textbf{Log Odds} = \ln\left(\frac{p(C_1|\mathrm{x})}{p(C_2|\mathrm{x})}\right) = \ln\left(\frac{p(C_1|\mathrm{x})}{1-p(C_1|\mathrm{x})}\right) = \mathrm{w}^\top \mathrm{x}$$

Algebra

$$\frac{p(C_1|\mathrm{x})}{1-p(C_1|\mathrm{x})} = \exp(\mathrm{w}^\top \mathrm{x})$$

$$p(C_1|\mathrm{x}) = \exp(\mathrm{w}^\top \mathrm{x})\left(1-p(C_1|\mathrm{x})\right)$$

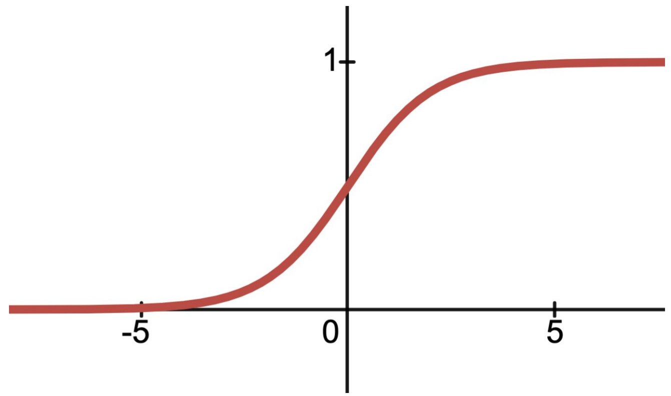
$$p(C_1|\mathrm{x}) = \exp(\mathrm{w}^\top \mathrm{x}) - \exp(\mathrm{w}^\top \mathrm{x})\, p(C_1|\mathrm{x})$$

$$p(C_1|\mathrm{x}) + \exp(\mathrm{w}^\top \mathrm{x})\, p(C_1|\mathrm{x}) = \exp(\mathrm{w}^\top \mathrm{x})$$

$$p(C_1|\mathrm{x}) = \frac{\exp(\mathrm{w}^\top \mathrm{x})}{1+\exp(\mathrm{w}^\top \mathrm{x})} \quad \boxed{= \frac{1}{1+\exp(-\mathrm{w}^\top \mathrm{x})}}$$

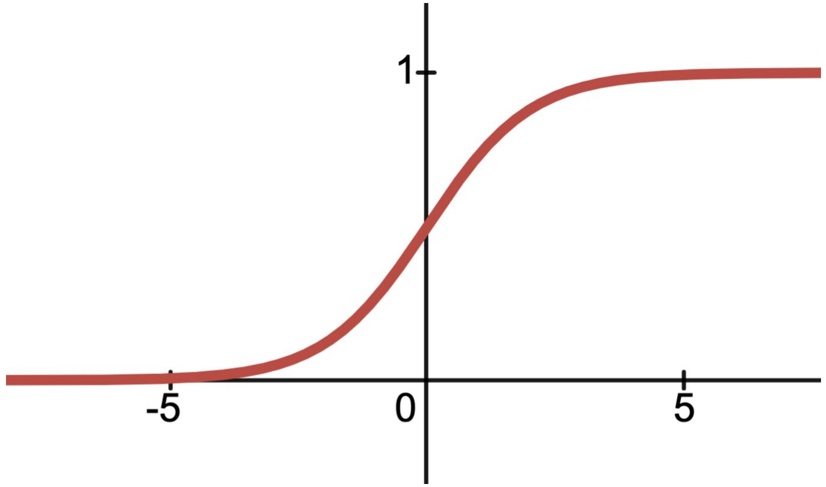
Sigmoid Function

$$\sigma(a) = \frac{1}{1+e^{-a}}$$



The Sigmoid (σ) Function

Also known as the **logistic** function.



$$\sigma(a) = \underbrace{\frac{1}{1+e^{-a}}}_{\text{sigmoid function}}$$

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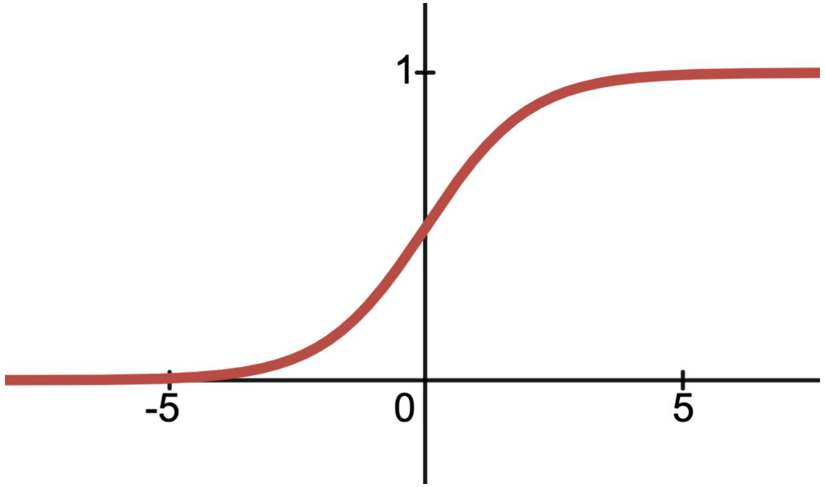


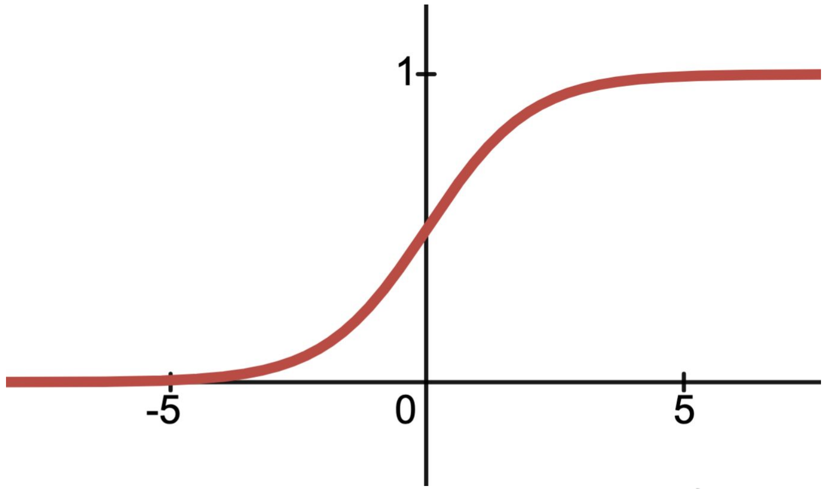
$$\underbrace{\sigma(a) = \frac{1}{1+e^{-a}}}_{\text{sigmoid function}}$$

The term Sigmoid means S-shaped.

A 'squashing function' since it maps the real axis into a finite interval.

Transforming the Sigmoid (σ) Function

We can **shift** and **scale** the sigmoid function just like any other function:



$$\sigma(Aa + B) = \frac{1}{1 + e^{-(Aa+B)}}$$

(Desmos)

Bigger A → Horizontal shrink. Steeper!
Bigger B → Negative horizontal shift

The Sigmoid (σ) Function

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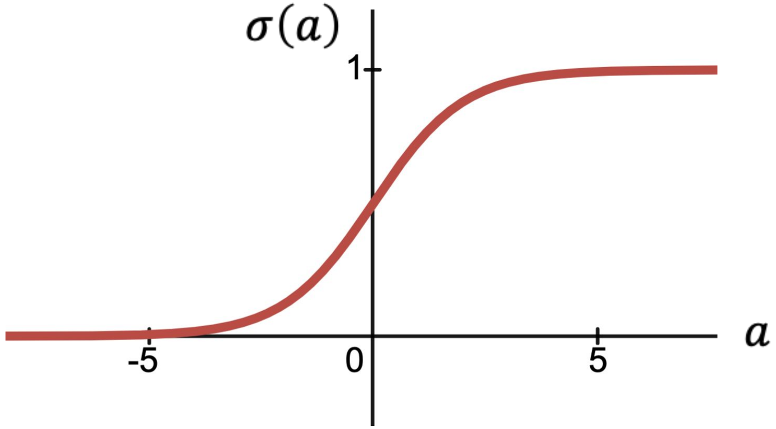
The S-shaped curve is formally known as the **sigmoid function**.

$$\sigma(a) = \frac{1}{1 + e^{-a}}$$



Reflection/ Symmetry

$$1 - \sigma(a) = \frac{e^{-a}}{1 + e^{-a}} = \sigma(-a)$$

Domain

$$-\infty < a < +\infty$$

Range

$$0 < \sigma(a) < 1$$

Derivative

$$\frac{d}{da}\sigma(a) = \sigma(a)(1 - \sigma(a)) = \sigma(a)\sigma(-a)$$

Inverse

$$a = \sigma^{-1}(p) = \log\left(\frac{p}{1 - p}\right)$$

Logit Function

Which of the following are properties of the sigmoid function, σ(x)?

Sigmoid for Classification

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Logistic Regression

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- Widely used models for **binary classification:**

  $x =$ "Get a FREE sample ..."

  $\phi(\mathrm{x}) = [2.0, 0, \ldots, 1.0, 0.5]$ ➡ $y = 1$

1 = "Spam"
0 = "Ham"

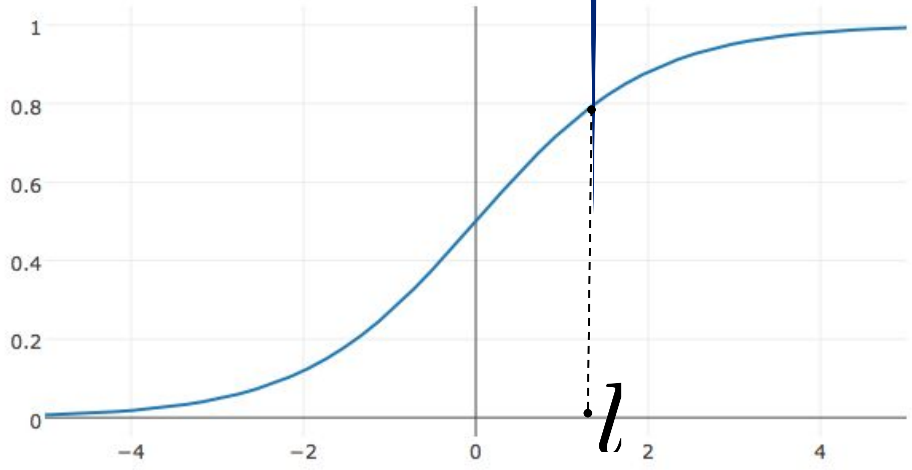
Why is ham good and spam bad? ... (https://www.youtube.com/watch?v=anwy2MPT5RE)

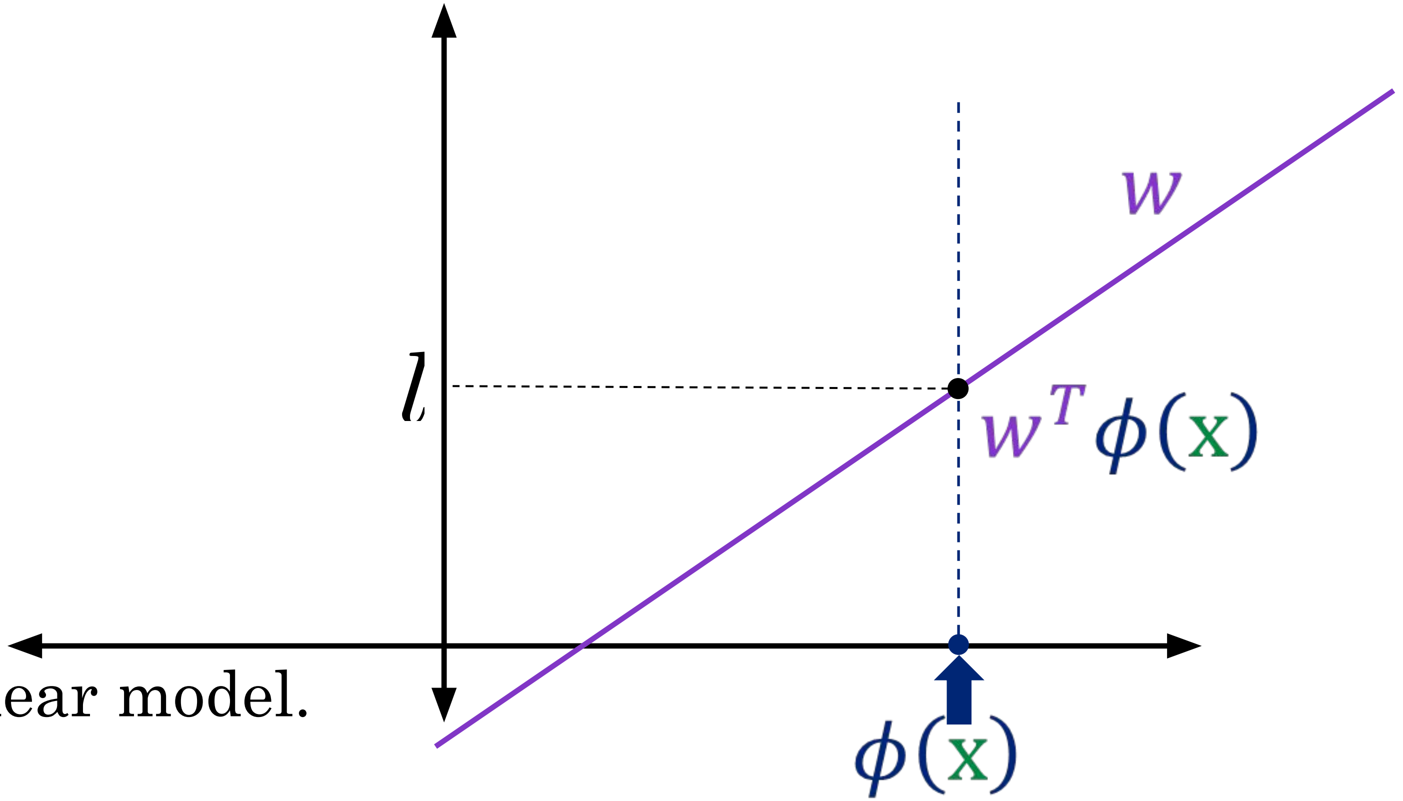
- Models the probability $p(C_1 | \phi(\mathrm{x}))$.

$$P(C_1|\phi(\mathrm{x})) = \sigma\left(w^T\phi(\mathrm{x})\right) = \frac{1}{1 + \exp\left(-w^T\phi(\mathrm{x})\right)}$$

Logistic Regression

Model: $P(C_1|\phi(\mathrm{x})) = \sigma\Big(\overbrace{w^T\phi(\mathrm{x})}^{\textbf{Linear Model}}\Big) = \dfrac{1}{1 + \exp\big(-w^T\phi(\mathrm{x})\big)}$





Generalized Linear Model:
Non-linear transformation of a linear model.

Logistic Regression

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- Widely used models for **binary classification:**

$$x = \text{“Get a FREE sample ...”}$$

$$\phi(\mathrm{x}) = [2.0, 0, \dots, 1.0, 0.5] \quad \Rightarrow \quad y = 1$$

1 = "Spam"
0 = "Ham"

Why is ham good and spam bad? … (https://www.youtube.com/watch?v=anwy2MPT5RE)

- Models the probability $p(C_1 | \phi(\mathrm{x}))$.

$$P(C_1|\phi(\mathrm{x})) = \sigma\left(w^T\phi(\mathrm{x})\right) = \frac{1}{1+\exp\left(-w^T\phi(\mathrm{x})\right)}$$

$$P(C_2|\phi(\mathrm{x})) = 1 - P(C_1|\phi(\mathrm{x}))$$

Our Cancer Classification Example

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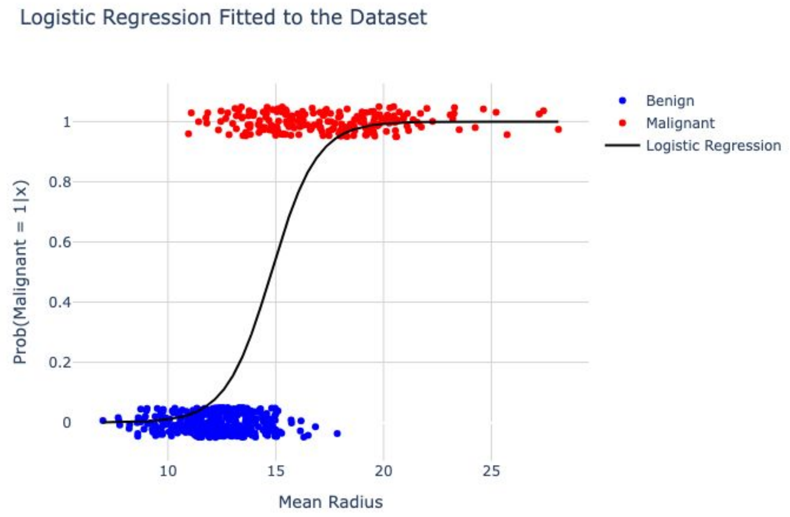
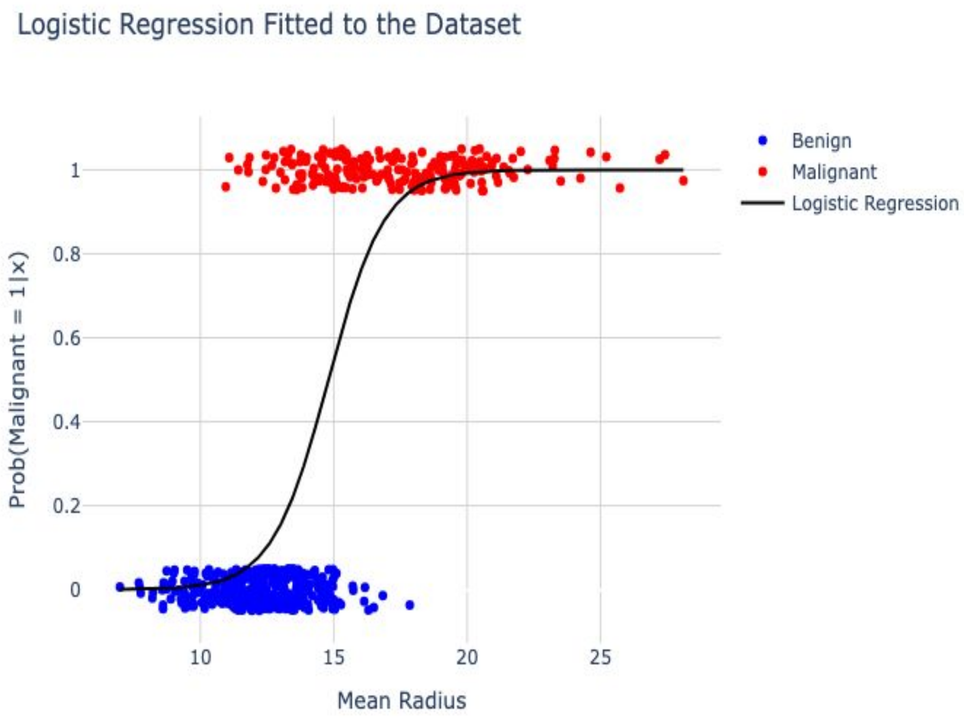
We've discovered the (simple) **logistic regression** model! It just fits a sigmoid to the data.

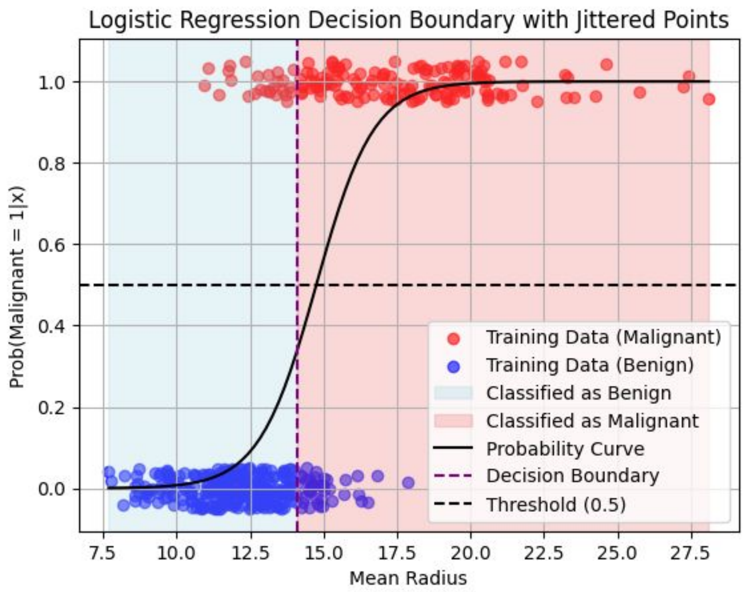


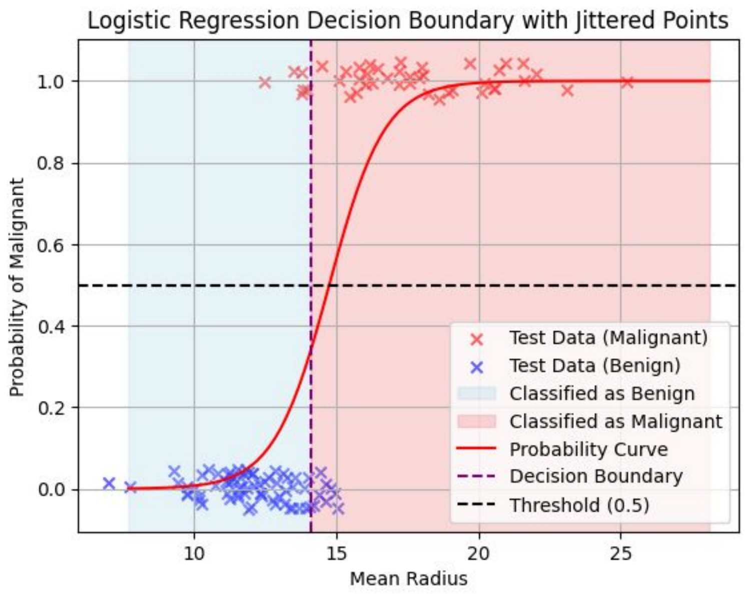
$$p(C_1|\phi(\mathrm{x})) = \sigma(w_0 + w_1\mathrm{x}) = \frac{1}{1 + e^{-(w_0 + w_1\mathrm{x})}}$$

How do we identify the optimal w's? More shortly!

Logistic Regression and Decision Boundary







Basis Functions and Decision Boundary

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- We can still apply basis functions $\phi(\mathrm{x})$ to the classification inputs. The resulting decision boundaries will be linear in the feature space ϕ, and these correspond to nonlinear decision boundaries in the original x.

Non-linear decision boundary in input space.

Linear decision boundary in the Gaussian space.

Centers of the Gaussian basis functions

Logistic Regression Objective

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Can We Use Squared Error As Objective?

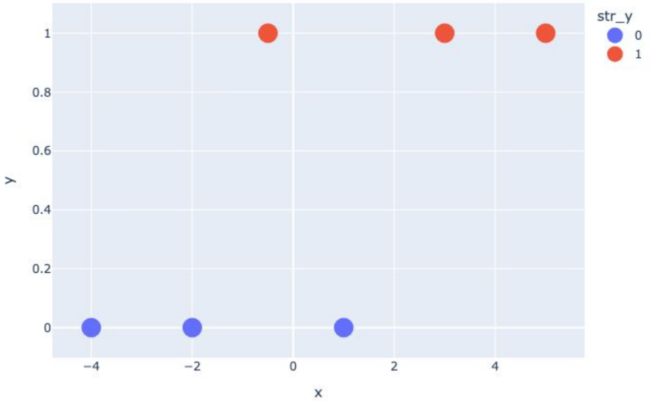
Toy Dataset: Squared Error

Logistic Regression model:

$$p(C_1|\phi(\mathrm{x})) = \sigma(w^T\phi(\mathrm{x}))$$

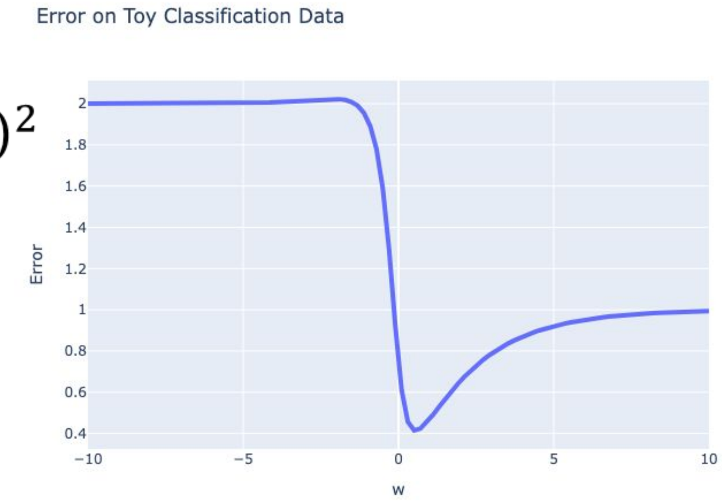
Assume no intercept.
So x and w are scalars.

	x	y
0	-4.0	0
1	-2.0	0
2	-0.5	1
3	1.0	0
4	3.0	1
5	5.0	1



Squared Error:

$$E(w) = \frac{1}{2}\sum_{n=1}^{N}(t_n - \sigma(w^T\phi(\mathrm{x})))^2$$



The squared error surface for logistic regression has many issues!

Pitfalls of Squared Error

1. Non-convex. Gets stuck in local minima.

$$E(w) = \frac{1}{2}\sum_{n=1}^{N}(t_n - \sigma(w^T\phi(\mathrm{x})))^2$$

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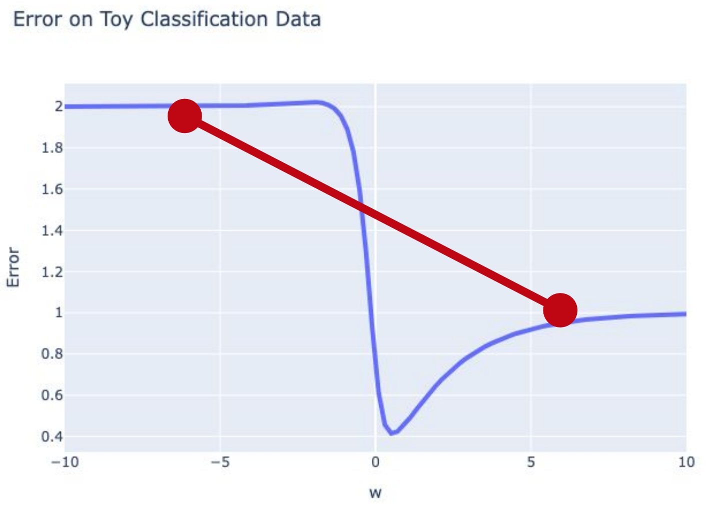
Secant line crosses function, so $E''(w)$ is not greater than 0 for all w.



Pitfalls of Squared Error

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$$E(w) = \frac{1}{2}\sum_{n=1}^{N}\left(t_n - \sigma(w^T\phi(\mathrm{x}))\right)^2$$

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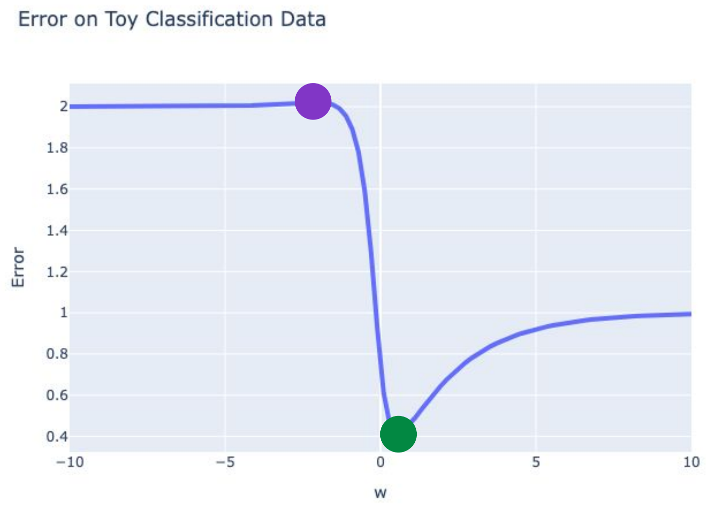


```
from scipy.optimize import minimize

minimize(ss_error_on_toy_data, x0 = 0)["x"][0]
```

0.544660182558169

```
minimize(ss_error_on_toy_data, x0 = -5)["x"][0]
```

-10.858380927026204

Pitfalls of Squared Error

1. Non-convex. Gets stuck in local minima.

$$E(w) = \frac{1}{2}\sum_{n=1}^{N}\left(t_n - \sigma(w^T\phi(\mathrm{x}))\right)^2$$

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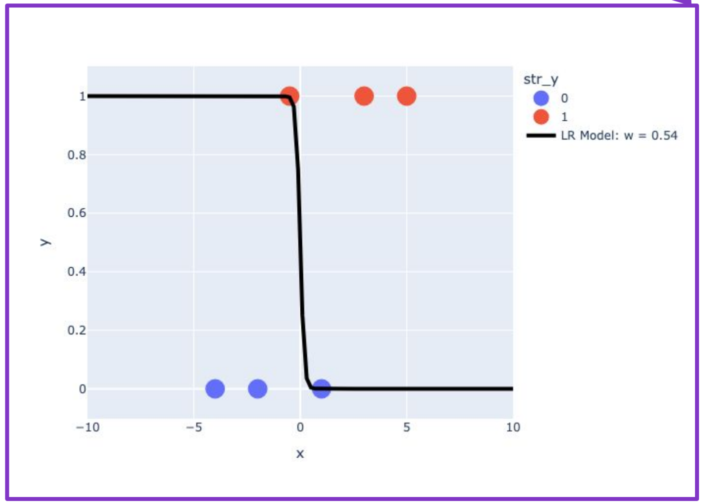

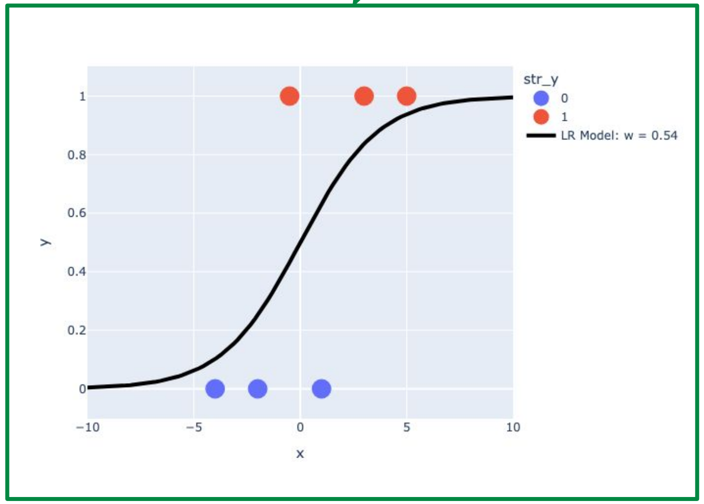

Squared Error Gives Bounded Penalty

- Best case scenario:

Estimate matches outcome.

Squared error = $(0-0)^2 = (1-1)^2 = \mathbf{0}$. Perfect prediction!

- Worst case scenarios:

Estimated probability = 0, Actual outcome = 1.

Squared error = $(0-1)^2 = \mathbf{1}$

Estimated probability = 1, Actual outcome = 0.

Squared error = $(1-0)^2 = \mathbf{1}$

- The largest possible squared error is 1. Not a huge penalty for a big mistake!

Pitfalls of Squared Error

$$E(w) = \frac{1}{2}\sum_{n=1}^{N}(t_n - \sigma(w^T\phi(\mathrm{x})))^2$$

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1. **Non-convex.** Gets stuck in local minima.

2. **Bounded.** Not a good measure of model error.

We'd like error functions to penalize "off" predictions. SE never gets very large, because both response and estimated probability are bounded by 1.

If true $t = 1$ but predicted probability = 0:

$$(t-p)^2 = (1-0)^2 = 1$$

A New Error Function for Logistic Regression

The Logistic Regression Objective (MLE)

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- In logistic regression, we can adopt **maximum likelihood estimation**.
- Define target variable $t \in \{0, 1\}$, with $t = 1$ representing class C_1 and $t = 0$ for class C_2.
- The likelihood of t given x under our model $p(C_1|\mathrm{x}) = \sigma(w^\top \mathrm{x})$:

$$p(t|w) = \underbrace{\sigma(w^\top \mathrm{x})^t}_{\text{for } t_n = \mathbf{1}\text{, only this term stays}} \underbrace{[1 - \sigma(w^\top \mathrm{x})]^{1-t}}_{\text{for } t_n = \mathbf{0}\text{, only this term stays}}$$

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$$p(t|w) = \sigma(w^\top \mathrm{x})^t \, [1 - \sigma(w^\top \mathrm{x})]^{1-t}$$

- For a dataset $\mathcal{D} = \{\mathrm{x}_n, t_n\}$, where $y_n = \sigma(w^\top \mathrm{x})$ the likelihood function can be written as:

$$p(\mathcal{D}|w) = \prod_{n=1}^{N} y_n^{t_n} (1 - y_n)^{1 - t_n}$$

The Logistic Regression Objective (MLE)

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$$p(\mathcal{D}|w) = \prod_{n=1}^{N} y_n^{t_n}(1 - y_n)^{1 - t_n}$$

We can define an error function by taking the **negative logarithm** of the **likelihood**, which gives the **cross-entropy error function**.

$$E(w) = -\ln p(\mathcal{D}|w) = -\sum_{n=1}^{N}[t_n \ln y_n + (1 - t_n)\ln(1 - y_n)]$$

Cross-Entropy: Two Error Functions In One!

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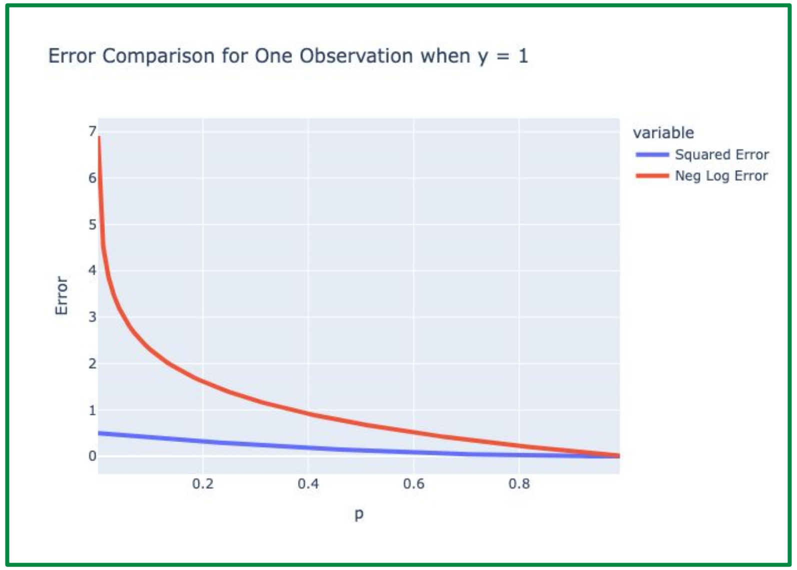
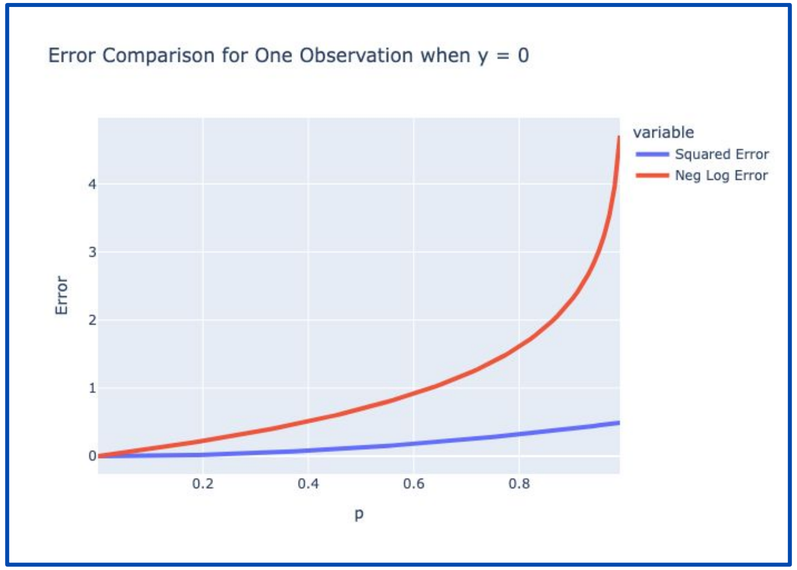
The Cross-Entropy (CE) function is often written more compactly as the **product of the label** and **the log probability**, summed over both classes.

$$\text{CE error} = \begin{cases} -\ln y_n\,, & \text{if } t_n = 1 \\ -\ln 1 - y_n & \text{if } t_n = 0 \end{cases}$$

$$-[\underbrace{t_n \ln y_n}_{\text{for } t_n = 1\text{, only this term stays}} + \underbrace{(1 - t_n)\ln(1 - y_n)}_{\text{for } t_n = 0\text{, only this term stays}}]$$





Unbounded for wrong-way certainty

What w value will achieve 0 cross-entropy error if we train on the datapoint x = 1, t = 1?

Convexity By Picture

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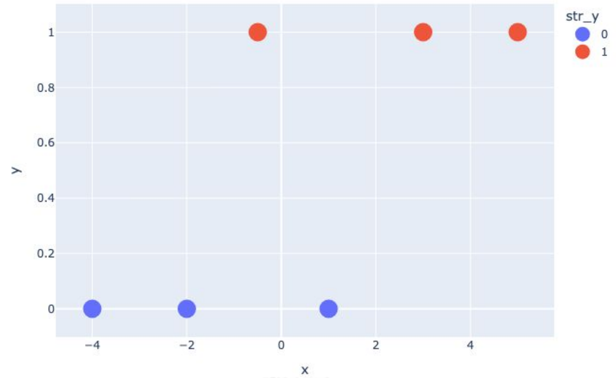
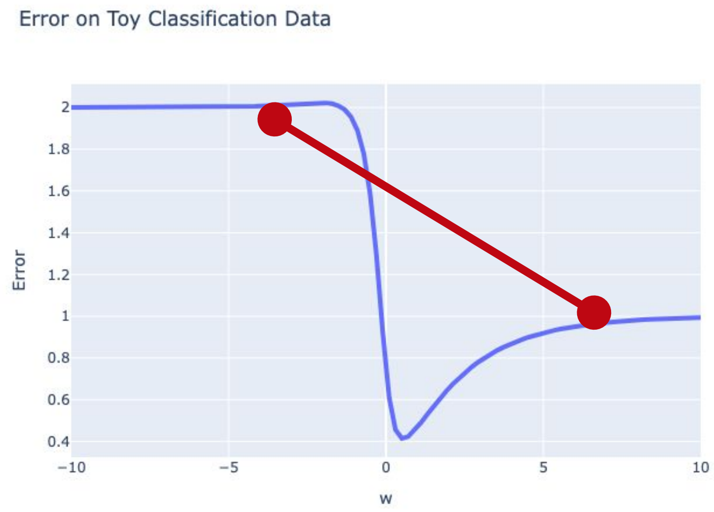
$$w = \underset{w}{\operatorname{argmin}} - \sum_{n=1}^{N} [t_n \ln y_n + (1 - t_n) \ln(1 - y_n)]$$



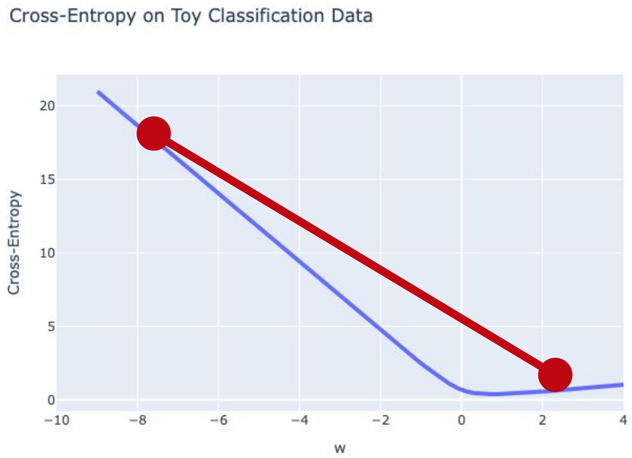
	x	y
0	-4.0	0
1	-2.0	0
2	-0.5	1
3	1.0	0
4	3.0	1
5	5.0	1

Squared Error Surface



A straight line crosses the curve Non-convex

Cross-Entropy Error Surface



Convex!

Regularization with Logistic Regression

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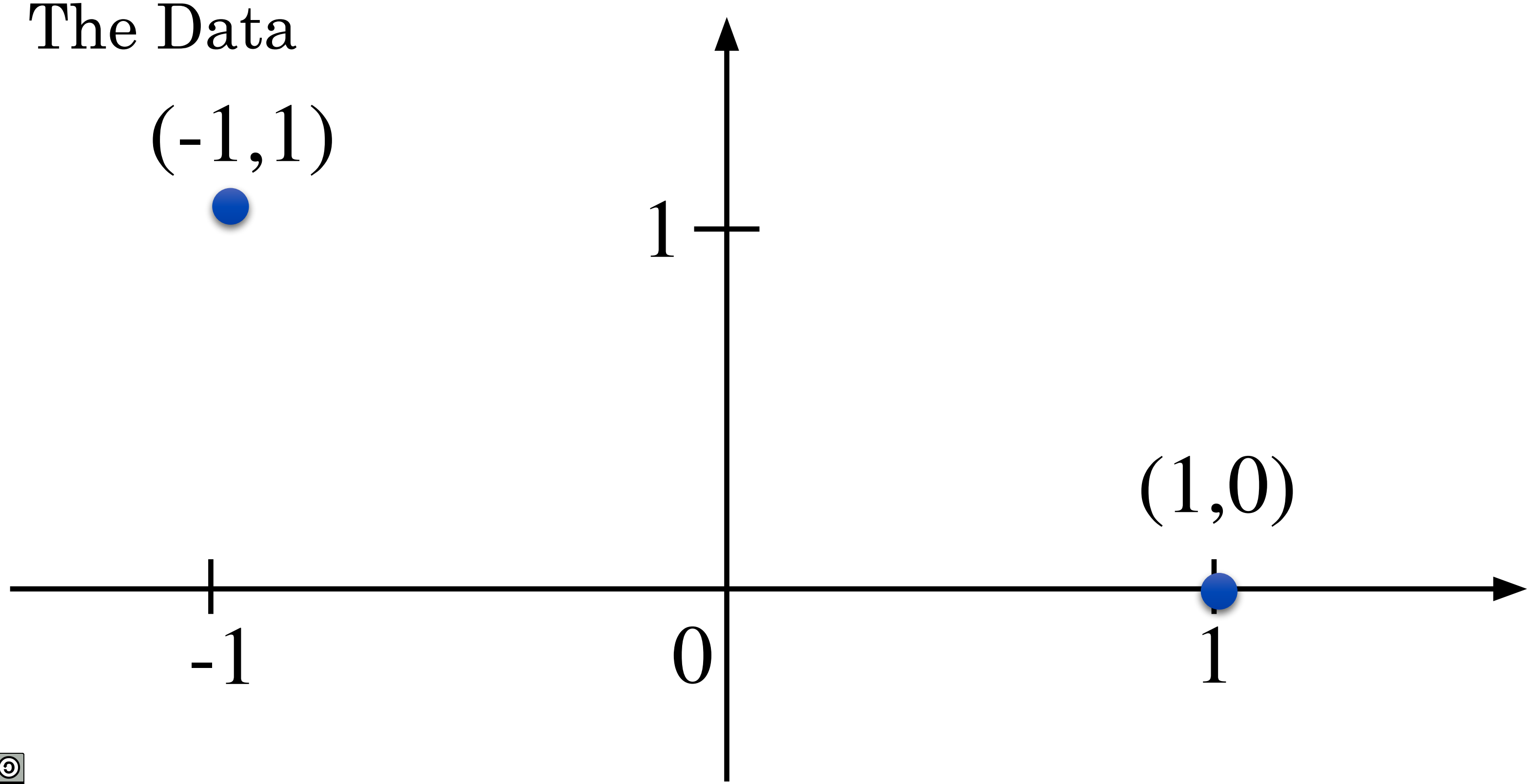
- Classification Task
- Defining a New Model for Classification
- Linear Discriminant Functions
- Discriminative Probabilistic Models
- Sigmoid for Classification
- Logistic Regression Objective
- Regularization with Logistic Regression
- Logistic Regression Optimization
- Making Decisions

What Is the Value of w?

$$w = \underset{w}{\operatorname{argmin}} - \sum_{n=1}^{N} [t_n \ln y_n + (1 - t_n) \ln(1 - y_n)]$$

Assume $\phi(\mathbf{x}) = \mathbf{x}$

The Data



A. $w = -1$

B. $w = 1$

C. $w = -\infty$

D. $w = +\infty$

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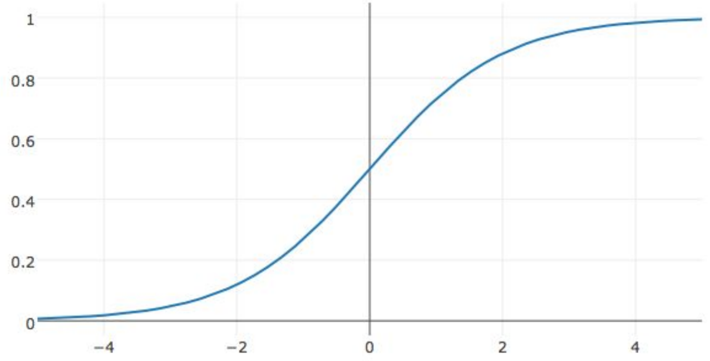
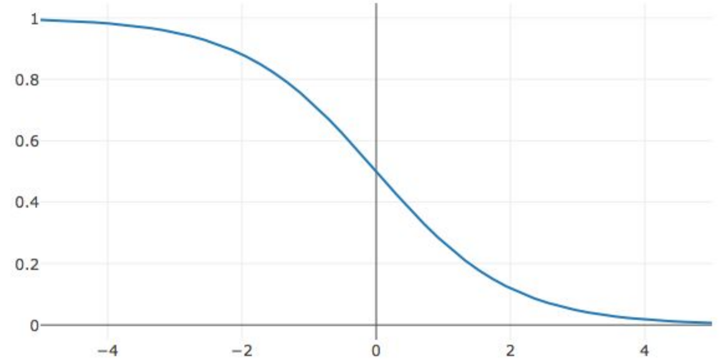




Do not edit
How to change the design

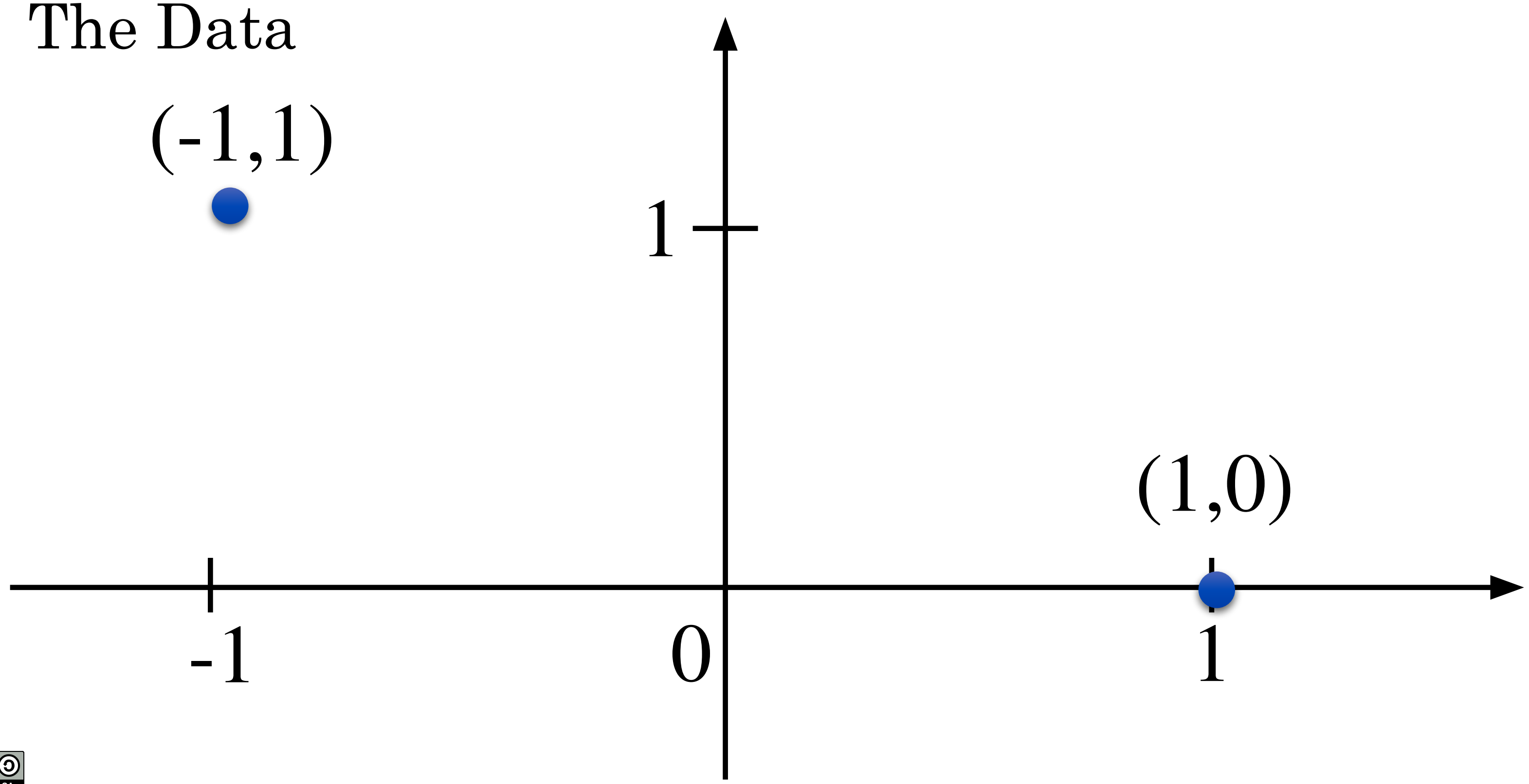
What is the best value of W?

What Is the Value of w?

$$w = \underset{w}{\operatorname{argmin}} - \sum_{n=1}^{N} [t_n \ln y_n + (1 - t_n) \ln(1 - y_n)]$$

Assume $\phi(\mathbf{x}) = \mathbf{x}$

The Data



A. $w = -1$

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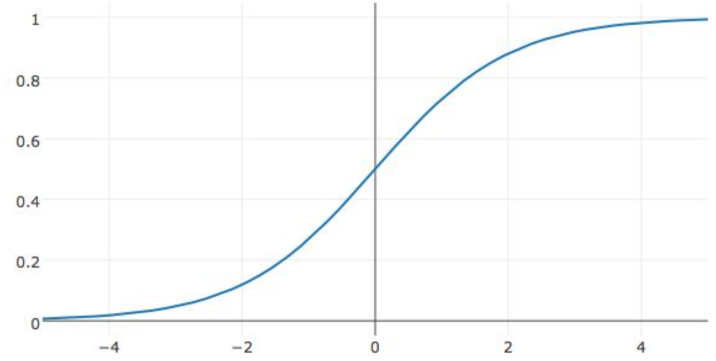

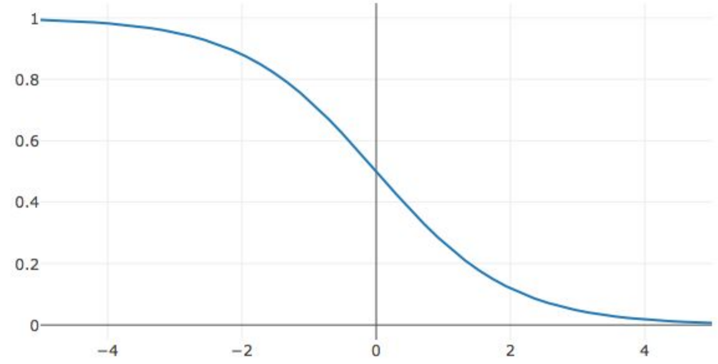

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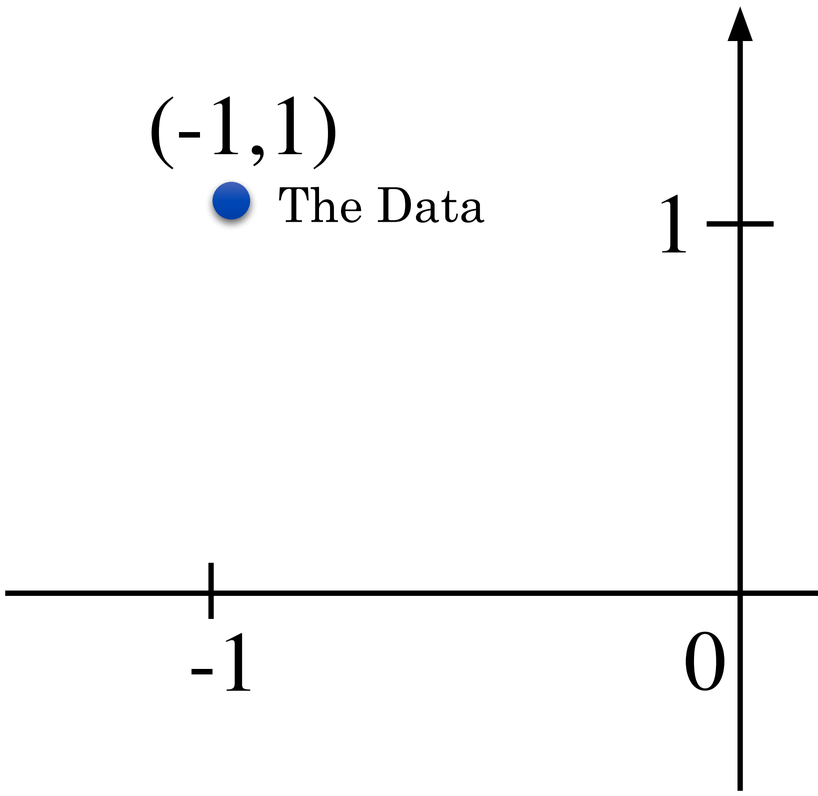
What Is the Value of w?

$$w = \underset{w}{\operatorname{argmin}} - \sum_{n=1}^{N} [t_n \ln y_n + (1 - t_n) \ln(1 - y_n)]$$

For the point (-1,1):

Objective:

$$-\ln \sigma(w^T \mathbf{x}) \quad = \quad -\ln \sigma(-w) \qquad \Rightarrow w \rightarrow -\infty$$



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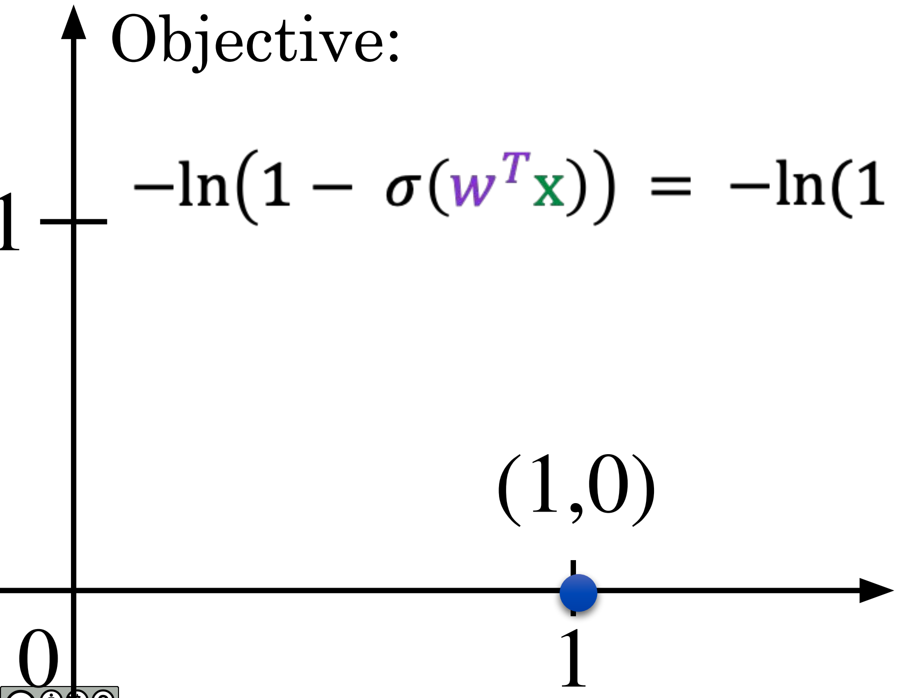
What Is the Value of w?

$$w = \underset{w}{\operatorname{argmin}} - \sum_{n=1}^{N} [t_n \ln y_n + (1 - t_n) \ln(1 - y_n)]$$

For the point (1,0):

Objective:

$$-\ln\left(1 - \sigma(w^T \mathbf{x})\right) = -\ln(1 - \sigma(w)) \quad \Rightarrow w \to -\infty$$



What Is the Value of w?

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For the point (-1,1):

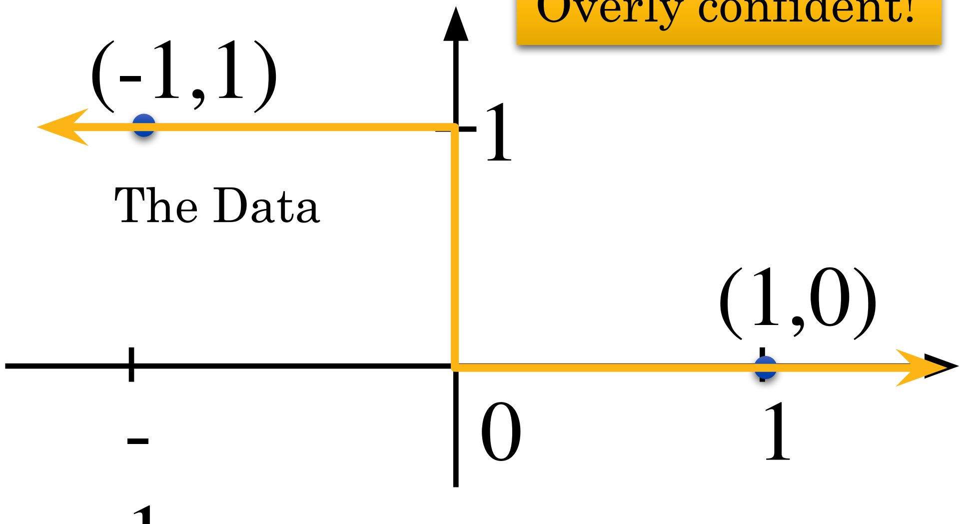
$$-\ln \sigma(w^T \mathbf{x}) \qquad \Rightarrow w \rightarrow -\infty$$

For the point (1,0):

$$-\ln\left(1 - \sigma(w^T \mathbf{x})\right) \qquad \Rightarrow w \rightarrow -\infty$$

Degenerate Solution!

Overly confident!

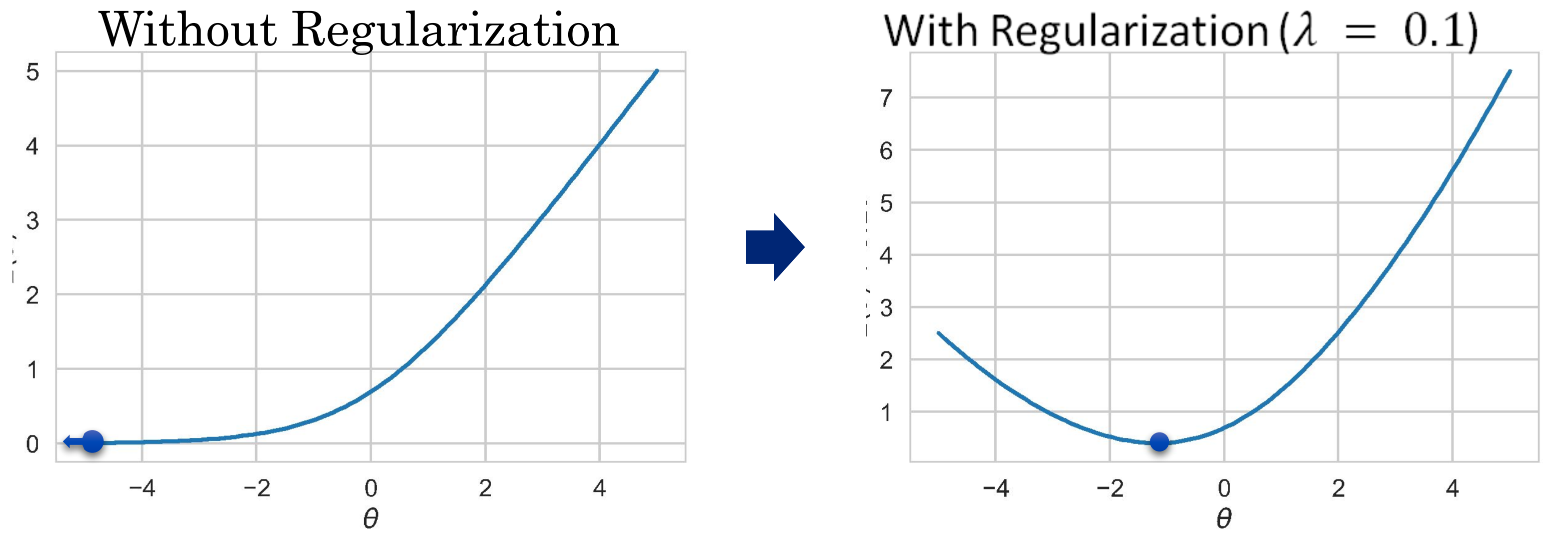


Adding Regularization to Logistic Regression

$$w = \underset{w}{\operatorname{argmin}} - \sum_{n=1}^{N} [t_n \ln y_n + (1 - t_n) \ln(1 - y_n)] + \frac{\lambda}{2} \sum_{d=1}^{D} w^2$$

Prevents weights from diverging on linearly separable data.

Earlier Example



Logistic Regression Optimization

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Logistic Regression Optimization

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$$E(w) = -\ln p(\mathbf{t}|w) = -\sum_{n=1}^{N}\left[\underbrace{t_n \ln y_n}_{\frac{t_n}{y_n}\frac{\partial y_n}{\partial w}} + \underbrace{(1 - t_n)\ln(1 - y_n)}_{\frac{(1 - t_n)}{1-y_n}\frac{\partial(1-y_n)}{\partial w}}\right]$$

Derivative wrt w using chain rule:

$$\frac{\partial E(w)}{\partial w} = -\sum_{n=1}^{N}\left[\frac{t_n}{y_n}\frac{\partial y_n}{\partial w} + \frac{(1 - t_n)}{1-y_n}\frac{\partial(1-y_n)}{\partial w}\right]$$

Logistic Regression Optimization

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$$E(w) = -\ln p(\mathbf{t}|w) = -\sum_{n=1}^{N}\Big[\underbrace{t_n \ln y_n}_{\frac{t_n}{y_n}\frac{\partial y_n}{\partial w}} + \underbrace{(1-t_n)\ln(1-y_n)}_{\frac{(1-t_n)}{1-y_n}\frac{\partial(1-y_n)}{\partial w}}\Big]$$

Derivative wrt w using chain rule:

$$\frac{\partial E(w)}{\partial w} = -\sum_{n=1}^{N}\left[\frac{t_n}{y_n}\frac{\partial y_n}{\partial w} + \frac{(1-t_n)}{1-y_n}\frac{\partial(1-y_n)}{\partial w}\right]$$

$$\boxed{\frac{\partial(1-y_n)}{\partial w} = -\frac{\partial y_n}{\partial w}}$$

$$= -\sum_{n=1}^{N}\left[\frac{t_n}{y_n}\frac{\partial y_n}{\partial w} - \frac{(1-t_n)}{1-y_n}\frac{\partial y_n}{\partial w}\right]$$

$$E(w) = -\ln p(\mathbf{t}|w) = -\sum_{n=1}^{N}[\underbrace{t_n \ln y_n}_{\frac{t_n}{y_n}\frac{\partial y_n}{\partial w}} + \underbrace{(1 - t_n)\ln(1 - y_n)}_{\frac{(1 - t_n)}{1-y_n}\frac{\partial(1-y_n)}{\partial w}}]$$

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Derivative wrt w using chain rule:

$$\frac{\partial E(w)}{\partial w} = -\sum_{n=1}^{N}\left[\frac{t_n}{y_n}\frac{\partial y_n}{\partial w} + \frac{(1 - t_n)}{1-y_n}\frac{\partial(1-y_n)}{\partial w}\right]$$

$$\boxed{\frac{\partial(1-y_n)}{\partial w} = -\frac{\partial y_n}{\partial w}}$$

$$= -\sum_{n=1}^{N}\left[\frac{t_n}{y_n}\frac{\partial y_n}{\partial w} - \frac{(1 - t_n)}{1-y_n}\frac{\partial y_n}{\partial w}\right]$$

$$= -\sum_{n=1}^{N}\left[\frac{t_n}{y_n} - \frac{(1 - t_n)}{1-y_n}\right]\frac{\partial y_n}{\partial w}$$

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$$E(w) = -\ln p(\mathbf{t}|w) = -\sum_{n=1}^{N} [t_n \ln y_n + (1 - t_n)\ln(1 - y_n)]$$

$$\frac{\partial E(w)}{\partial w} = -\sum_{n=1}^{N}\left[\frac{t_n}{y_n}\frac{\partial y_n}{\partial w} + \frac{(1 - t_n)}{1 - y_n}\frac{\partial(1 - y_n)}{\partial w}\right]$$

$$\boxed{\frac{\partial(1 - y_n)}{\partial w} = -\frac{\partial y_n}{\partial w}}$$

$$= -\sum_{n=1}^{N}\left[\frac{t_n}{y_n}\frac{\partial y_n}{\partial w} - \frac{(1 - t_n)}{1 - y_n}\frac{\partial y_n}{\partial w}\right]$$

$$= -\sum_{n=1}^{N}\left[\frac{t_n}{y_n} - \frac{(1 - t_n)}{1 - y_n}\right]\boxed{\frac{\partial y_n}{\partial w}}$$

$$y_n = \sigma(w^T\phi(\mathrm{x}_n))$$

Need to apply chain rule

$$\frac{\partial E(w)}{\partial w} = -\sum_{n=1}^{N}\left[\frac{t_n}{y_n} - \frac{(1 - t_n)}{1 - y_n}\right] y_n(1 - y_n)\phi(\mathrm{x}_n)$$

$$\frac{\partial y_n}{\partial w} = \frac{\partial \sigma(w^T\phi(\mathrm{x}_n))}{\partial w} = \sigma(w^T\phi(\mathrm{x}_n))\left(1 - \sigma(w^T\phi(\mathrm{x}_n))\right)\phi(\mathrm{x}_n) = y_n(1 - y_n)\phi(\mathrm{x}_n)$$

$$E(w) = -\ln p(\mathbf{t}|w) = -\sum_{n=1}^{N} [t_n \ln y_n + (1 - t_n) \ln(1 - y_n)]$$

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$$\frac{\partial E(w)}{\partial w} = -\sum_{n=1}^{N} \left[\frac{t_n}{y_n} - \frac{(1 - t_n)}{1 - y_n}\right] y_n(1 - y_n)\phi(\mathrm{x}_n)$$

Separating the terms

$$= -\sum_{n=1}^{N} [t_n(1 - y_n)\phi(\mathrm{x}_n) - (1 - t_n)y_n\phi(\mathrm{x}_n)]$$

$$= -\sum_{n=1}^{N} [t_n\phi(\mathrm{x}_n) - \cancel{t_n y_n \phi(\mathrm{x}_n)} - y_n\phi(\mathrm{x}_n) + \cancel{t_n y_n \phi(\mathrm{x}_n)}]$$

$$= -\sum_{n=1}^{N} [t_n\phi(\mathrm{x}_n) - y_n\phi(\mathrm{x}_n)] = \sum_{n=1}^{N} [y_n - t_n]\phi(\mathrm{x}_n)$$

We Cannot Solve Directly

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$$E(w) = -\ln p(\mathbf{t}|w) = -\sum_{n=1}^{N}[t_n \ln y_n + (1 - t_n)\ln(1 - y_n)]$$

$$\frac{\partial E(w)}{\partial w} = \sum_{n=1}^{N}[y_n - t_n]\phi(\mathrm{x}_n)$$

$$= \sum_{n=1}^{N}[\sigma(w^T\phi(\mathrm{x}_n)) - t_n]\phi(\mathrm{x}_n)$$

No Closed Form Solution

w is inside the σ for every term so we cannot pull out w.

We cannot apply logit to inverse σ:
$logit\,(a + b) \neq logit(a) + logit\,(b)$

More on this in lectures 12 and 13 with Gradient Descent.

Making Decisions

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Decisions = Posteriors + Loss

- We separate ***inference*** from ***decision***.

  Inference gives posteriors $p(C_k|\mathrm{x})$.

  The task gives a **loss matrix** L_{kj}.

$\boldsymbol{L_{kj}}$: Loss if we predict j when the true class is k.

The right action is the one with **minimum expected loss at this** x.

$$\text{Expected loss at } \mathrm{x} \text{ if we choose class } j = \sum_k L_{kj}\, p(C_k|\mathrm{x})$$

Input x → **Model outputs** → Decision computes → Pick

Decisions = Posteriors + Loss

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The right action is the one with **minimum expected loss at this** x.

Expected loss at x if we choose class j $= \sum_k L_{kj} p(C_k|\mathrm{x})$

Input x → **Model outputs** → Decision computes → Pick

Change the **costs** L → decisions change immediately;
No retraining. That's why we prefer probabilities over hard labels.

Expected Loss Example

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In the example of cancer prediction, let's assume for a data sample we have:
$p(C_0|\mathrm{x}) = 0.7$ and $p(C_1|\mathrm{x}) = 0.3$.

Assume also that the Loss matrix is given as:

$$\begin{array}{ccc} & C_0 & C_1 \\ & \text{normal} & \text{cancer} \\ \begin{array}{c}\text{normal}\\ \text{cancer}\end{array} & \multicolumn{2}{c}{\begin{pmatrix} 0 & 1 \\ 100 & 0 \end{pmatrix}} \end{array}$$

If we predict normal
$j = C_1$

$$\sum_{k=C_0} L(j,k)p(C_1|\mathrm{x}) = 100 \times 0.3 = 30$$

If we predict cancer
$j = C_0$

$$\sum_{k=C_1} L(j,k)p(C_0|\mathrm{x}) = 1 \times 0.7 = 0.7$$

$0.7 < 30$ and predicting cancer gives smaller loss so we predict the class of cancer.

The Rejection Option

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- Reject decisions when max posterior probability ≤ threshold θ.
- Control error rate vs. rejection fraction.

Lecture 10

Logistic Regression (1)

Credit: Joseph E. Gonzalez and Narges Norouzi

Reference Book Chapters: Chapter 5 Sections 5.1, 5.2(up to 5.2.5), 5.4 (up to 5.4.4)